A Latent Oscillator Measurement Model to Simulate Emotional-Expression Score Dynamics in Video

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Abstract

Facial-expression classifiers convert video into multivariate time series of scores with measurement error from classifiers, videos, and recording conditions. Empirical score series cannot establish whether the score channels reflect a smaller set of latent expressive processes or whether an analysis would recover those processes. We introduce the Latent Oscillator Measurement Model (LOMM), a data-generating model that separates latent dynamics, time-varying activity, and a factor-analytic observation model. The latent processes are damped, undamped, or amplifying linear oscillators. LOMM generates bounded scores or continuous indicators. Study 1 used four-fold cross-fitting with scores from 100 MAFW videos to calibrate LOMM and evaluate generated series on held-out videos. Median plausibility and coverage were 0.970 and 0.920 for LOMM, versus 0.510 and 0.370 for a calibrated static logistic-normal generator. Study 2 tested whether Dynamic Exploratory Graph Analysis (DynEGA), static EGA, GraphicalVAR, and GIMME recovered a known dimensional structure from continuous indicators generated by LOMM. At 100 observations per clip, with failed or timed-out fits counted as incorrect, correct-dimension recovery was 0.939 for DynEGA, 0.884 for static EGA, 0.777 for GraphicalVAR, and 0.176 for GIMME. Replacing the common fixed initialization with independent stationary starts for stable dimensions and bounded independent starts for amplifying dimensions reduced recovery for DynEGA, static EGA, and GraphicalVAR. The primary initialization was optimistic relative to this alternative. LOMM provides a controlled test of whether an analysis recovers a specified latent structure before score patterns are interpreted psychologically.

Introduction

Facial-expression classifiers convert video of a person into multivariate time series. Each frame yields a vector of scores, one for each label in a chosen label set, at a temporal resolution that manual coding cannot match.

Earlier classifiers returned scores only for the categories in their training data. Vision-language models such as CLIP can score any label supplied as text (Foteinopoulou & Patras, 2023; Li et al., 2023; Radford et al., 2021). The scores are nonetheless measurements: they depend on the model, the label set, the video-processing pipeline, image quality, and recording conditions, and their accuracy varies with model adaptation, label choice, and evaluation data (Foteinopoulou & Patras, 2023; Li et al., 2023). In a small demonstration on FindingEmo images, zero-shot CLIP achieved 41.7% accuracy and a macro-F1 of 0.331 (Tomašević et al., 2026). On the Image-Emotion dataset, fine-tuned Emotion-CLIP reached 67% accuracy and a macro-F1 of 0.64 in a reported 10-fold evaluation (Bondielli & Passaro, 2021). A pattern among score channels may therefore reflect expressive behavior, the measurement process, or both. In this paper, classifier scores are treated as fallible measurements of visible expression, not as measures of subjective emotional experience.

This raises a psychometric question. Score channels can be analyzed as separate variables, or they can be treated as fallible indicators of a smaller number of latent expressive processes (Mehu & Scherer, 2015; Nesselroade et al., 2002; Russell, 2003). The number of classifier labels does not establish how many distinct processes the channels represent. Treating overlapping channels as separate dimensions can overstate that number, while combining indicators of distinct processes can conceal differences. Dimensionality assessment therefore concerns both the number of latent processes and the assignment of channels to those processes.

The observable associations among indicators can change even when the generating dimensions and indicator assignments remain fixed. Dynamics, activity, measurement error, and observation length affect how clearly the dimensional structure appears in a recorded series. Indicators of the same process may share little observable variation when that process is weakly expressed or measurement error is large, and measurement error can change the structure recovered by dynamic methods (Kankaanpää et al., 2026). Changes in activity can also produce changes in the indicators without a discontinuity in the underlying latent level. Evaluating recovery therefore requires conditions in which these sources of variation can be controlled while the generating structure remains known.

We introduce the Latent Oscillator Measurement Model (LOMM) for this purpose. LOMM combines the damped linear oscillator model of intraindividual dynamics (Boker, 2015; Boker & Nesselroade, 2002) with an activity layer and a factor-analytic observation model (Nesselroade et al., 2002; Zhang et al., 2008). It simulates latent processes, specifies how strongly each process is expressed at each observation, and maps the expressed processes onto fallible indicators. These components and the observation length can be varied while the generating dimensions and indicator assignments remain fixed, allowing researchers to examine when a specified structure is recovered and which conditions obscure it. Empirical score series can inform dimensionality judgments and simulation settings; their unknown generating structure limits direct checks

of recovery.

We use LOMM in two studies (Figure 1). In Study 1 the MAFW classifier-score series are the target (Liu et al., 2022). Each of four folds uses 75 MAFW videos to calibrate LOMM and 25 videos to evaluate the generated series. A static logistic-normal generator provides a comparison without temporal dependence. Eleven statistics computed on scores and log-ratio coordinates define series-level plausibility and coverage. In Study 2 the LOMM design is the target. LOMM generates continuous indicators with a known number of dimensions and a known assignment of indicators to dimensions, and four methods estimate a network among the indicators and group them into communities: Dynamic Exploratory Graph Analysis (DynEGA), static EGA, GraphicalVAR, and GIMME. The question is whether the recovered communities match the structure that generated the data. Static EGA ignores temporal order and is computationally inexpensive; it shows how much structure is recoverable from contemporaneous indicator levels alone. Study 2 also examines two limits of the main result. One analysis compares the common fixed initialization with independent stationary starts for stable dimensions and bounded independent starts for amplifying dimensions. A second tests whether the indicator grouping is preserved after the indicators are transformed to bounded scores that sum to one, as classifier scores do. Because MAFW videos contain a median of 35 scored observations, although uninterrupted segments are shorter, with a median length of 13.5, Study 2 also includes a $T = 25$ shorter-record stress test.

The Latent Oscillator Measurement Model

LOMM represents a standard psychometric situation: several observed variables are fallible indicators of a smaller number of latent processes, and those processes change over time. As an illustration, suppose a simulation has three latent dimensions labeled approach-affiliation, threat-defense, and orienting-novelty, and the anger, fear, disgust, and anxiety channels are assigned as indicators of threat-defense. Those four channels would covary over time because they load on the same latent dimension, and each would also carry channel-specific measurement error. A recovery method should group the four channels together and should find three dimensions. These labels only illustrate a possible simulation design; whether MAFW scores have this structure is not tested here.

LOMM generates data in three layers (Figure 2). The latent layer describes how each latent dimension changes over time. The activity layer determines how strongly that dimension is expressed at each observation. The observation layer creates multiple indicators with prespecified loadings and measurement error. Keeping these layers separate allows a researcher to change the dynamics, the expression pattern, or the measurement

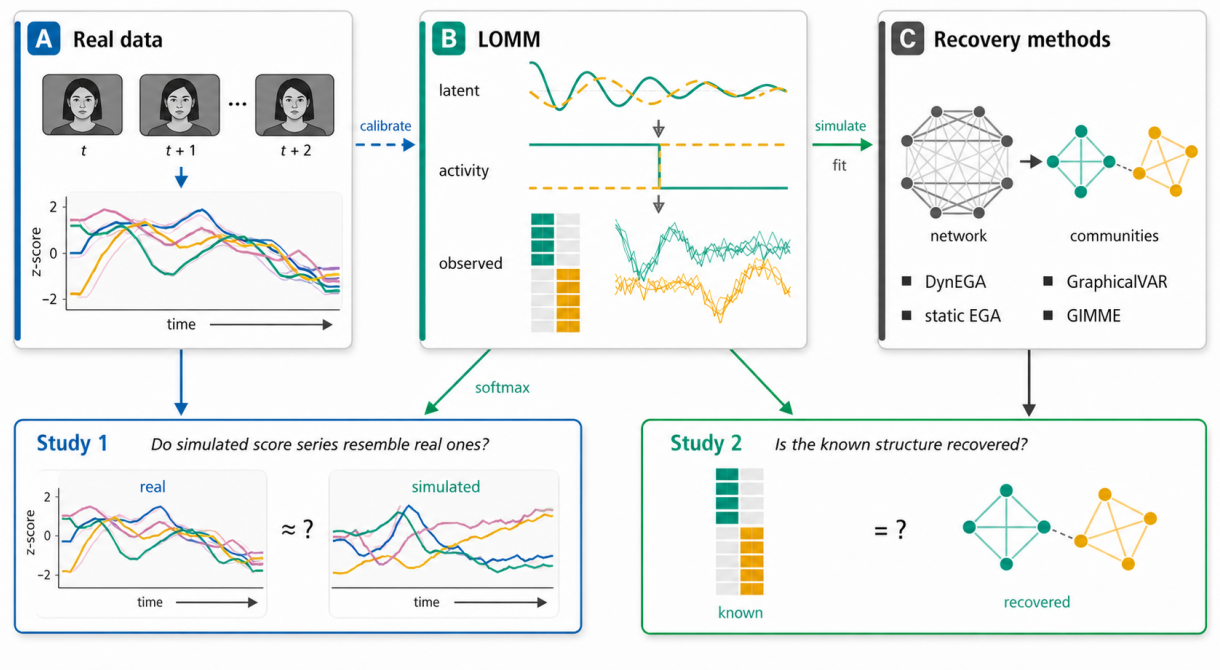


Figure 1: Overview of the two studies. (A) Real data: a classifier scores each sampled frame of an MAFW video, giving one score series per label. (B) LOMM: latent oscillators, an activity schedule that sets how strongly each is expressed at each observation, and observed indicators with a known assignment to dimensions (colored blocks). (C) Recovery methods: DynEGA, static EGA, GraphicalVAR, and GIMME each estimate a network among the indicators and group them into communities. Study 1 uses MAFW calibration videos to set the generator and compares simulated score series with MAFW evaluation series. Study 2 fits the four methods to simulated indicators and compares the recovered communities with the known assignment.

quality while holding the other components fixed.

Notation. Clips are indexed by $i = 1, \dots, N$ and observations within a clip by $n = 1, \dots, T$, at times $t_n = (n-1)\Delta$, where Δ is the observation interval. Latent dimensions are indexed by $f = 1, \dots, m$. Continuous indicators are indexed by $j = 1, \dots, p$; in Study 2, $p = mk$ with k indicators per dimension. Bounded scores have J channels. Bold symbols are vectors or matrices.

For clip i and latent dimension f, let $x_{if}(t)$ denote the latent level and $\dot{x}_{if}(t)$ its rate of change. LOMM treats each dimension as a linear second-order oscillator:

$$\ddot{x}_{if}(t) = \eta_f x_{if}(t) + \zeta_f \dot{x}_{if}(t) + q_{if,n}, \qquad t \in [t_n, t_{n+1}). \tag{1}$$

Equation 1 is the damped linear oscillator used in research on intraindividual dynamics (Boker & Nesselroade, 2002). The position-feedback parameter η_f determines the strength of the restoring force toward equilibrium; more negative values produce faster oscillation. The velocity-feedback parameter ζ_f determines damping: negative values damp the motion, zero leaves it undamped, and positive values amplify it. The term $q_{if,n}$ is a random acceleration input, drawn independently for each interval between observations n and $n+1$ and held constant within that interval; it produces irregular peaks and reversals in the trajectory. When η_f and ζ_f are both negative, the trajectory returns toward equilibrium, with oscillation when the characteristic roots are complex. Positive velocity feedback produces growing amplitudes. We use these configurations as finite-record stress tests and report their realized amplitudes in Appendix C. They are not stationary models of sustained expressive behavior. Appendix A gives the exact discrete-time transition used to advance the system between observations, the stability conditions, and the initialization rules.

The activity layer multiplies the latent level at each observation by a nonnegative activity value, $\tilde{x}_{if,n} = a_{if,n} x_{if,n}$. A latent dimension can therefore remain strongly expressed throughout a clip, remain near a low background level, or alternate with other dimensions. This layer matters for recovery because an abrupt change in activity can produce a large change in the observed indicators even when the latent level remains continuous.

The observation layer is a standard factor model:

$$\mathbf{z}_{in} = \Lambda_i^{(z)} \tilde{\mathbf{x}}_{in} + \varepsilon_{in}^{(z)}, \qquad \varepsilon_{in}^{(z)} \sim \mathcal{N}(\mathbf{0}, \Psi_i). \tag{2}$$

Row j of $\Lambda_i^{(z)}$ contains the loadings of indicator j on the m latent dimensions, and $\varepsilon_{in}^{(z)}$ is measurement error

with covariance Ψ_i. Appendix A gives the complete observation equations, including the separate notation for continuous indicators and log-ratio coordinates and the optional clip-specific baseline and scale terms.

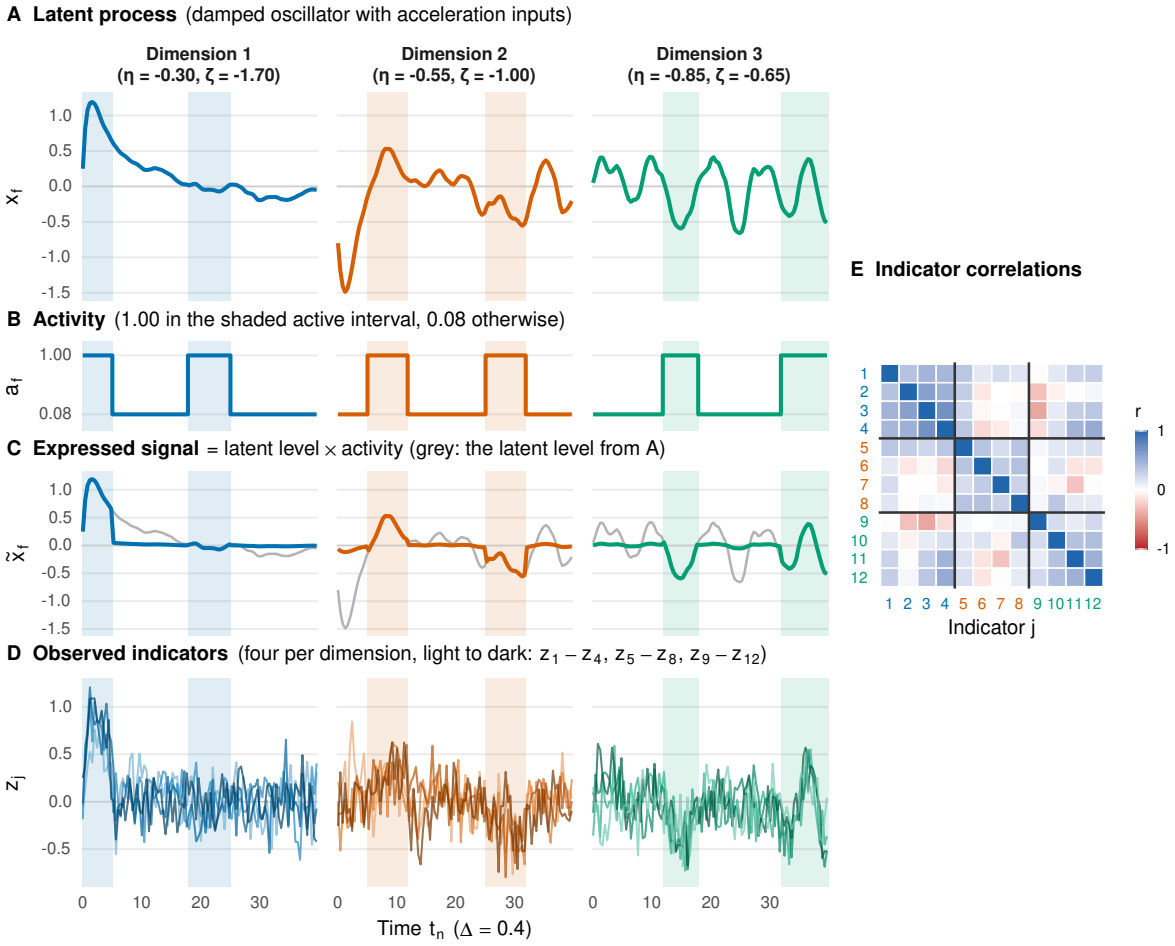


Figure 2: One simulated LOMM clip with three latent dimensions, $T = 100$ observations, and $\Delta = 0.4$. Each column follows one dimension from the latent layer to the observed indicators. (A) Latent trajectory; the oscillator parameters are given in the column headers, and each dimension has its own initial state and acceleration inputs. (B) Activity, 1.00 in the shaded active interval and 0.08 otherwise. (C) Expressed signal, the product of the latent level and the activity value, drawn over the latent level in grey. (D) The four observed indicators of each dimension in shades from light to dark, with primary loadings 0.65 to 0.85, secondary loadings of ± 0.30 or ± 0.35 on another dimension for nine of the twelve indicators, and residual SD 0.20. (E) Correlations among the twelve indicators from this clip, ordered by dimension.

Figure 2 shows the three layers for one simulated clip. In panel A the dimensions differ in their feedback parameters and start from different states: dimension 1 rises to an early peak and decays smoothly, dimension 2 starts in a trough and rebounds, and dimension 3 oscillates faster and is driven mainly by its acceleration inputs. Panel B shows the activity schedule, in which the dimensions take turns being active. Panel C shows the product of the two layers. A latent peak that falls inside an active interval is expressed at full size; the same peak outside the interval is reduced to 8% of its size and is barely visible in the indicators. The steps at

the boundaries of the active intervals appear in the indicators although the latent level remains continuous. Panel D shows the twelve indicators; each loads mainly on its own dimension, and nine also carry a smaller loading on another dimension. Panel E shows the correlation matrix among the twelve indicators from this clip. The three diagonal blocks reflect the indicator assignments, which is the structure that Study 2 asks the methods to recover. The weaker correlations between blocks can reflect both secondary loadings and sample covariation among the expressed latent trajectories. The secondary loadings in this example are larger than in the Study 2 cross-loading condition, which limits them to ± 0.10.

LOMM can output two types of observed variables. The first is the vector of continuous indicators $\mathbf{z}_{in}$, used in the primary Study 2 designs; there, the factor structure that generated the data is defined directly on the analyzed variables. The second is a vector of J positive scores that sum to one, which matches the form of classifier output and is used in Study 1. To produce bounded scores, LOMM maps the latent dimensions to a $(J-1)$-dimensional vector of log-ratio coordinates and applies the softmax function. Because scores that sum to one are compositional data, they are analyzed both as scores and in log-ratio coordinates, which remove the sum constraint. A separate Study 2 condition applies the softmax directly to the p continuous indicators and tests whether their original grouping is preserved. Appendix A gives the softmax, log-ratio, and identification details.

In Study 1, MAFW calibration videos determine the LOMM observation-model geometry and generator settings. The primary Study 2 design sets between-clip baselines to zero and scales to one but lets loadings and residual variances vary slightly across clips. Estimating oscillator, activity, and measurement parameters from empirical data would require the sign, scale, temperature, and frequency constraints stated in Appendix A.

Study 1: Comparing Simulated and Real Score Series

Study 1 examined whether LOMM, after calibration to MAFW, could generate classifier-score series with statistical properties observed in MAFW videos excluded from calibration. The analysis used 100 videos from MAFW, a large multimodal database of facial-expression video recorded in naturalistic settings (Liu et al., 2022). Frames were scored with the transforEmotion R package (Tomašević et al., 2026) using OpenAI's CLIP ViT-Large/14 model (setting `oai-large`), with up to 50 frames sampled per video. Eleven labels from the MAFW label set served as text prompts: anger, disgust, fear, happiness, neutral, sadness, surprise, contempt, anxiety, helplessness, and disappointment. For each scored frame, the eleven classifier-score channels sum to one.

Figure 3 shows the classifier-score series for one MAFW video. Frames without a detected face have missing scores, so each video was divided into segments of consecutive scored frames. The analysis contains 3,557 observations in 188 segments. Videos contribute 16 to 50 scored observations, with a median of 35; segment lengths range from 1 to 50 observations, with a median of 13.5. Appendix B gives the face-detection, score-validity, missing-data, and sampling details.

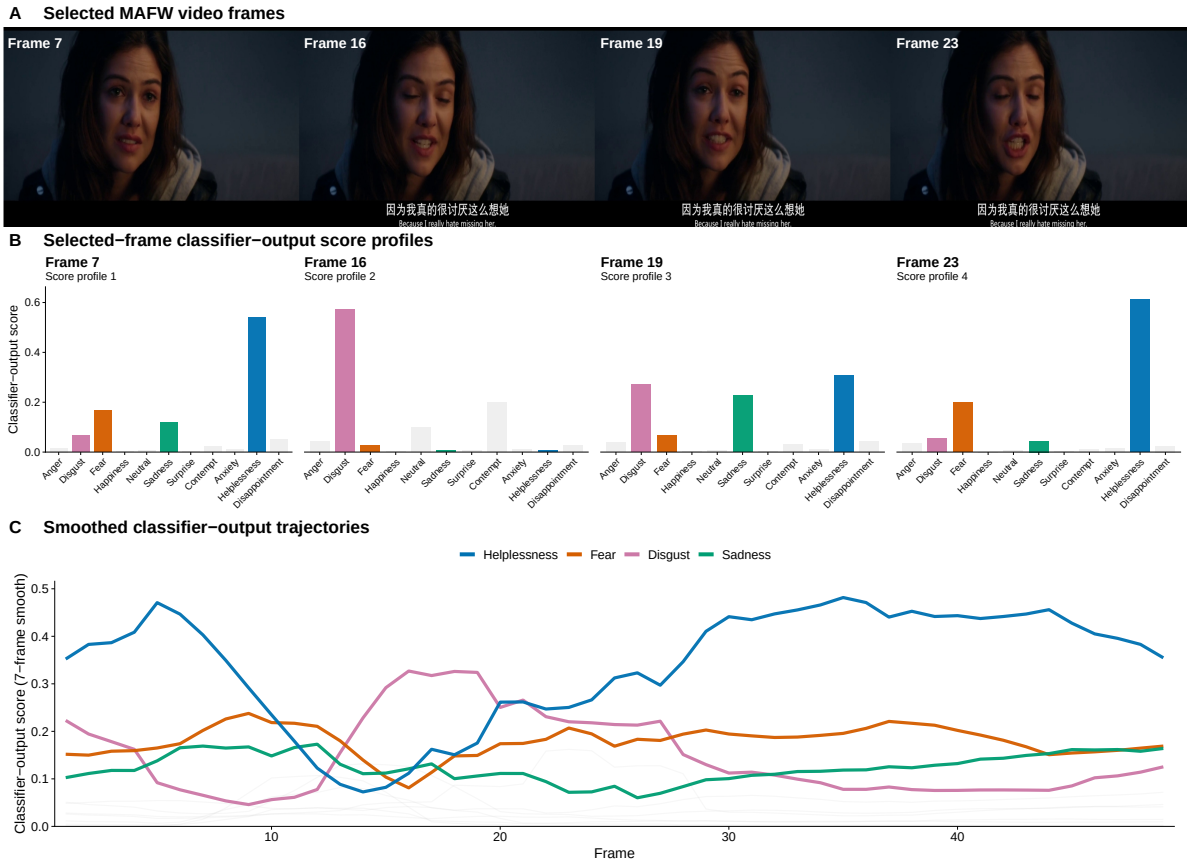


Figure 3: Classifier-score series from one MAFW video. Panel A shows four frames from the video, reproduced under the MAFW research-use terms. Panel B shows the eleven classifier scores for those frames. Panel C shows smoothed trajectories for the channels with the highest mean scores. Scores were generated with transforEmotion using OpenAI CLIP ViT-Large/14.

We represented each MAFW and simulated series with a broad panel of statistics covering location, dispersion, short-term change, temporal persistence, temporal irregularity, and cross-channel dimensionality. Eight statistics were computed within channel and video: median level; interquartile range of levels; interquartile range of observation-to-observation changes; lag-1 autocorrelation; and run-length, sample, permutation, and spectral entropy. Their channel-specific values were summarized by the median within each video. Effective rank, robust level variance, and robust change variance described all channels jointly. Each statistic was evaluated on the classifier scores and a log-ratio representation, producing 22 features. Channel statistics

used centered log-ratios (CLR), and the three joint statistics used isometric log-ratio (ILR) coordinates. The main analysis combines the complete feature panel in the plausibility and coverage measures. Appendix C reports the metric-specific comparisons.

We used four-fold cross-fitting. In each fold, 75 MAFW videos formed the calibration set and 25 formed the evaluation set. The calibration set determined the LOMM measurement geometry and the dynamic and measurement parameters. LOMM then generated 100 datasets with the same observation positions and uninterrupted segment boundaries as the evaluation videos. We compared LOMM with a static logistic-normal generator calibrated from the same 75 MAFW videos. The static generator retained the estimated compositional structure, between-video heterogeneity, and observation pattern while drawing observations independently over time. This comparison estimates the gain associated with temporal generation beyond static calibration.

For each fold, the MAFW reference comprised 1,000 comparisons between two disjoint, category-stratified groups of 25 videos drawn from the 75 calibration videos. The 22 features were standardized using medians and robust scales estimated from the calibration set. Series similarity was measured by the root-mean-square Euclidean distance across these standardized features. The similarity threshold was the 95th percentile of the directional nearest-neighbor distances in the MAFW reference comparisons. Plausibility is the proportion of simulated series whose nearest MAFW evaluation series fell within this threshold. Coverage is the proportion of MAFW evaluation series whose nearest simulation fell within the threshold. Reported ranges are the 2.5th and 97.5th percentiles across simulation or reference replicates. Appendix B gives the calibration, parameter selection, metric, and reference-sampling procedures.

Study 2: Recovery of Dimensionality and Indicator Assignments

Data-generating conditions. Study 2 asks whether four network-based methods recover a prespecified number of dimensions and a prespecified assignment of indicators to dimensions. The primary analyses use continuous indicators generated by LOMM. Each dataset contains N clips with T observations each and $p = mk$ indicators. All clips in a dataset share the same indicator assignments and nominal measurement parameters. In the matched-method and extended factorial designs, each primary loading receives an independent $\mathrm{Unif}(-0.15, 0.15)$ deviation and each residual variance receives an independent $\mathrm{Unif}(-0.01, 0.01)$ deviation. Datasets are generated under six scenario families (Table 1), which differ in how strongly and how persistently each latent dimension is expressed and in whether the oscillator dynamics are stable or amplifying. Appendix B gives the activity schedules and implementation details; Table B2 gives the numerical oscillator

parameters.

Every method receives the same simulated dataset. Each method estimates its own structure, that structure is converted to an undirected network by a prespecified rule, and the same community-detection procedure identifies groups of indicators. The estimated grouping is then compared with the assignment used to generate the data. This creates a common recovery target while preserving each method's own statistical model and data handling.

Table 1: The six Study 2 scenario families. Numeric activity gains and dwell fractions are in Appendix B; oscillator parameters are in Table B2.

Code	Family
A	Weak expression, one dominant dimension
B	Strong expression, one dominant dimension
C	Strong expression, switching dominant dimension
D	Strong expression, rapid switching
E	Amplifying dynamics, one dominant dimension
F	Amplifying dynamics, switching dominant dimension

Table 2: The four Study 2 methods, the structure each supplies for the common grouping step, and its role in the comparison.

Method	Structure used for grouping	Role in the comparison
DynEGA	Partial-correlation network of estimated changes	Method of primary interest
Static EGA	Partial-correlation network of contemporaneous levels	Ignores temporal order; computationally inexpensive benchmark
GraphicalVAR	Contemporaneous residual partial-correlation network	Dynamic network model with lagged and contemporaneous relations
GIMME	Average residual-covariance network across clip-level fits	Fitted per clip and averaged; tests computational feasibility

Recovery procedures. DynEGA (Boker et al., 2010; H. Golino et al., 2020; H. Golino et al., 2022; H. F. Golino & Epskamp, 2017) is the method of primary interest. It estimates the first derivative of each indicator from time-delay embeddings, estimates a regularized partial-correlation network among the derivatives, and applies a community-detection algorithm to the network; the communities are the estimated dimensions.

GraphicalVAR (Epskamp et al., 2018; Epskamp, 2024) fits a multilevel vector autoregression that separates lagged from contemporaneous relations; its contemporaneous residual partial-correlation network is used for grouping. GIMME (Gates & Molenaar, 2012; Lane et al., 2019) fits a structural vector autoregression to each clip separately; residual covariance matrices from successful fits are averaged across clips, and the average is used for grouping. This use of GIMME is computationally demanding and differs from its intended idiographic application, so the GIMME results apply to this comparison only.

Static EGA estimates a regularized partial-correlation network from the contemporaneous correlation matrix of indicator levels, ignoring the observation order, and applies the same community-detection step. It serves two purposes. Statistically, it shows how much of the grouping is recoverable from contemporaneous levels alone; if most of it is, a dynamic model cannot add much for this recovery target. Computationally, it requires far less time than the dynamic methods. Comparing static EGA with DynEGA and GraphicalVAR therefore shows whether modeling temporal structure improves grouping enough to justify its computational cost in these conditions.

Primary design and outcomes. The primary design uses $T = 100$ observations per clip and contains 21,600 datasets. It crosses the six scenario families, $m \in \{2, 3\}$ dimensions, $N \in \{50, 100\}$ clips, three loading levels (0.40, 0.60, 0.80), three residual standard deviations (0.125, 0.25, 0.50), and 100 replications, with $k = 4$ indicators per dimension. Every method receives the same dataset within a condition and replication. Because MAFW videos contain a median of 35 scored observations, although uninterrupted segments are shorter, with a median length of 13.5, we generated 21,600 corresponding datasets with $T = 25$ as a shorter-record stress test. Louvain at resolution 1.0 is the primary community-detection algorithm; Leiden and Walktrap were applied to the same fitted networks as sensitivity analyses.

The primary outcome is the overall correct-dimension recovery rate: the proportion of datasets in which the method returned a valid network and the estimated number of dimensions equaled m. A failed or timed-out fit counts as an unsuccessful recovery. We also report the valid-output rate and the correct-dimension rate among valid outputs, so that computational feasibility and accuracy can be examined separately. Assignment accuracy, the adjusted Rand index, and variation of information compare the full estimated partition of indicators with the true partition among valid outputs. Secondary diagnostics assess alignment between the recovered groups and the activity-scaled latent trajectories using best-match absolute trajectory correlation and trajectory NRMSE. Appendix B defines how these diagnostics are constructed and matched.

Additional analyses. Three additional analyses examine the limits of the primary result. First, one-at-a-time sensitivity analyses start from a reference condition ($N = 100$, $T = 100$, loading 0.60, residual SD 0.25,

common fixed initialization) and alter one design component at a time by changing the initialization rule, smoothing activity transitions, adding small cross-loadings, correlating residuals, correlating acceleration inputs, or transforming the indicators to bounded scores. The alternative initialization uses independent stationary starts for stable dimensions and bounded independent starts for amplifying dimensions. The bounded-score condition applies the softmax function directly to the p continuous indicators and analyzes either the p scores or their centered log-ratio (CLR) components. Because the transformed scores must sum to one, the original block structure among the p variables need not be preserved. Second, a small analysis adds clip-specific baseline and scale differences, with baselines constrained to the column space of the loading matrix so that they add no structure outside the factor model. Third, an extended factorial simulation of 518,400 datasets evaluates DynEGA alone over additional numbers of dimensions, indicators per dimension, and observation schedules. Appendix B gives the exact settings for every analysis.

Results

Study 1 Results: Similarity of Simulated and Real Score Series

Figure 4 compares the two generators with the MAFW reference. LOMM achieved a median plausibility of 0.970 (95% simulation range [0.935, 0.990]) and coverage of 0.920 [0.900, 0.950]. The corresponding MAFW-to-MAFW reference values were 0.950 [0.910, 0.990] for both measures. The static logistic-normal generator achieved plausibility of 0.510 [0.410, 0.590] and coverage of 0.370 [0.290, 0.455]. LOMM therefore generated score series that usually had a close MAFW counterpart and represented most MAFW evaluation series, whereas static calibration alone covered substantially less of the empirical variation.

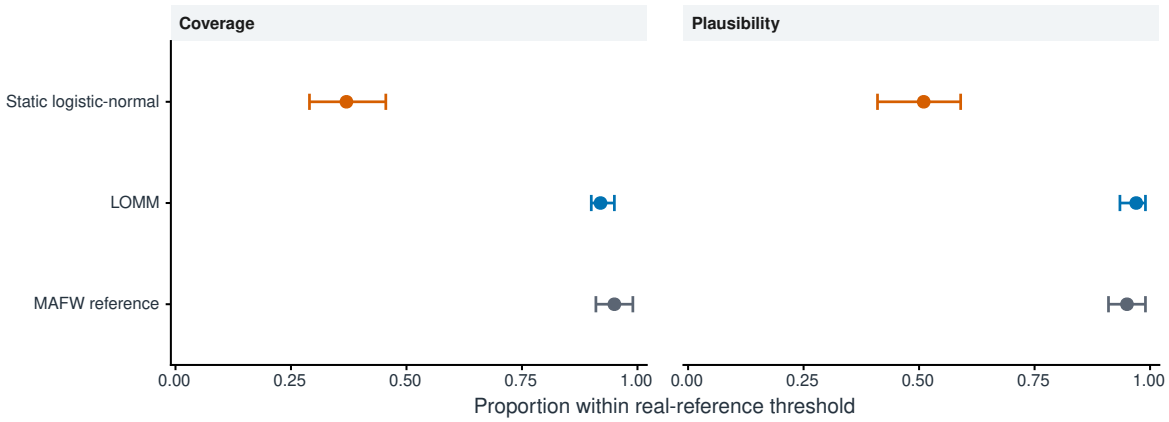


Figure 4: Cross-fitted similarity of LOMM and static logistic-normal series to MAFW evaluation series. Points are median proportions and bars are 2.5th and 97.5th percentiles across 100 generated datasets for each generator or 1,000 MAFW-to-MAFW reference comparisons. Plausibility is the proportion of simulated series with a nearest MAFW evaluation series within the fold-specific threshold. Coverage is the proportion of MAFW evaluation series with a nearest simulation within the same threshold. The threshold is the 95th percentile of directional nearest-neighbor distances between disjoint MAFW calibration samples. Higher values indicate closer agreement.

Across the metric-specific comparisons, 14 of the 22 LOMM feature discrepancies fell within the MAFW-to-MAFW reference range. Agreement was strongest for dispersion, short-term change, temporal persistence, and robust change variance. The clearest remaining differences concerned sample entropy, which was higher in LOMM series, and spectral entropy, which was lower. Appendix C reports all metric-specific results.

Study 2 Results: Recovery of Dimensionality and Indicator Assignments

Table 3 reports the $T = 100$ results under Louvain. The overall recovery rate requires both a valid network and the correct number of dimensions, so every failure and timeout counts against it. The last two columns report assignment accuracy among valid outputs and median runtime.

Table 3: Results for the primary $T = 100$ design under Louvain. Assignment accuracy is computed among valid network outputs. Runtime is the median, in seconds, among valid network outputs and includes model fitting, network extraction, and community detection.

Method	Valid output	Correct given valid	Overall recovery	Assignment accuracy	Median runtime, s
DynEGA	1.000	0.939	0.939	0.965	3.69
Static EGA	0.997	0.887	0.884	0.922	2.15
GraphicalVAR	0.941	0.826	0.777	0.902	85.41
GIMME	0.224	0.783	0.176	0.877	518.16

DynEGA returned a valid network for all 21,600 datasets and had the highest overall recovery rate, 0.939. Static EGA followed at 0.884, which shows that most of the grouping was recoverable from contemporaneous

indicator levels. GraphicalVAR reached 0.777. GIMME recovered the correct dimension in 0.783 of its valid outputs but returned a valid network for only 0.224 of datasets, so its overall recovery was 0.176. Assignment accuracy among valid outputs was 0.965 for DynEGA, 0.922 for static EGA, 0.902 for GraphicalVAR, and 0.877 for GIMME. Static EGA was about 40 times faster than GraphicalVAR and had higher overall recovery and assignment accuracy for this target. GraphicalVAR estimates a more complex model that includes lagged effects; the comparison evaluates only whether that additional structure improved recovery of the grouping under these conditions. Appendix C reports Monte Carlo standard errors, bootstrap intervals, and the remaining partition diagnostics.

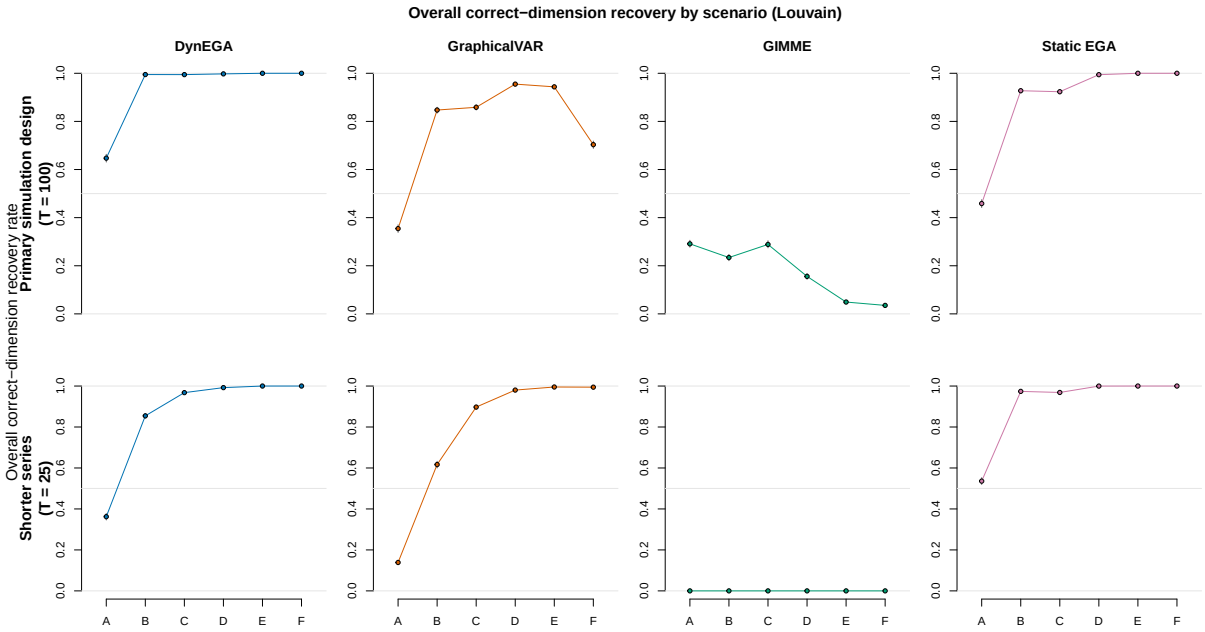


Figure 5: Overall correct-dimension recovery under Louvain by scenario family (codes as in Table 1), method, and observations per clip ($T = 100$, top; $T = 25$, bottom). A recovery requires a valid network output and the correct number of dimensions. Error bars are 95% bootstrap intervals from resampling whole datasets.

Recovery varied across scenario families and observations per clip (Figure 5). DynEGA had the highest recovery in most conditions. Static EGA's recovery was high because the block structure was present in contemporaneous levels. GraphicalVAR's recovery was lowest when expression was weak or measurement error was large. GIMME's low overall rates mainly reflected missing valid outputs. In the extended factorial simulation, DynEGA recovered the correct number of dimensions in 93.7% of 518,400 datasets and assigned 95.2% of indicators correctly; stronger loadings and smaller residual error improved recovery. Leiden and Walktrap led to the same conclusions as Louvain.

Secondary diagnostics assessed alignment between the recovered groups and the activity-scaled latent trajectories. Among valid outputs with the correct number of dimensions at $T = 100$, the best-match absolute trajectory correlation and trajectory NRMSE were 0.760 and 0.602 for DynEGA, 0.791 and 0.568 for GraphicalVAR, 0.606 and 0.898 for GIMME, and 0.787 and 0.563 for static EGA. GraphicalVAR and static EGA had higher trajectory correlations and lower NRMSE than DynEGA, while DynEGA had higher correlation and lower NRMSE than GIMME. These descriptive comparisons condition on the outputs satisfying each method's dimensional recovery criterion. Grouping recovery and trajectory recovery are different targets, and neither ordering ranks the methods in general. Appendix C reports the complete diagnostics.

Robustness and Sensitivity Analyses

Shorter records. With $T = 25$, static EGA had the highest overall recovery, 0.913, followed by DynEGA, 0.863, and GraphicalVAR, 0.770. Shortening the record slightly reduced DynEGA's recovery, left GraphicalVAR's about the same, and increased static EGA's. At $T = 25$, all 21,600 GIMME attempts ended in errors and none returned a valid network; none was recorded as a timeout. Among outputs with the correct number of dimensions, DynEGA's trajectory correlation was 0.848 and NRMSE was 0.521; GraphicalVAR had the highest correlation and lowest NRMSE, followed by DynEGA and static EGA (Appendix C).

Initial states. In the reference condition of the one-at-a-time analyses ($N = 100$, $T = 100$, loading 0.60, residual SD 0.25, common fixed initialization), overall recovery was 0.993 for DynEGA, 0.912 for static EGA, and 0.683 for GraphicalVAR. Replacing the common fixed initialization with independent stationary starts for stable dimensions and bounded independent starts for amplifying dimensions produced the largest change among the sensitivity conditions: recovery fell to 0.838, 0.525, and 0.253, respectively. The common fixed initialization was therefore optimistic relative to this alternative.

Other one-at-a-time changes. Gradual activity transitions, small cross-loadings, and correlated acceleration inputs changed recovery little. A residual correlation of 0.10 between every pair of indicators increased recovery for GraphicalVAR and static EGA. This result shows that the particular residual pattern made the target grouping easier to recover; it does not establish robustness to residual dependence more generally.

Bounded scores. Transforming the p indicators to scores that sum to one changed the variables being analyzed and sharply reduced recovery of the original grouping. DynEGA recovered the correct number of dimensions in 0.333 of datasets analyzed as CLR components and 0.233 analyzed as raw scores; the corresponding rates were 0.333 and 0.167 for static EGA and 0.087 and 0.002 for GraphicalVAR. The softmax transformation and the accompanying projection onto the zero-sum subspace did not preserve the structure among the original

indicators. For this reason, the Study 2 recovery results apply to continuous indicators and are not extended to bounded scores of the kind analyzed in Study 1.

Between-clip baseline and scale. This analysis used stable families A through D with independent stationary initial states. Its reference recovery rates are lower than those in analyses using the common fixed initialization. Adding clip-specific baseline and scale changed DynEGA's recovery from 0.775 to 0.750, a paired difference of -0.025 (95% interval $[-0.100, 0.050]$). Static EGA rose from 0.275 to 1.000: the added baselines lie in the column space of the loadings, so they add between-clip variation that follows the intended grouping. GraphicalVAR and GIMME were unchanged, but those estimates rest on very few valid outputs; GraphicalVAR produced three correct outputs among 80 datasets, and every three-dimensional GIMME fit timed out.

Runtime and feasibility. Median runtime among valid $T = 100$ outputs was 2.15 s for static EGA, 3.69 s for DynEGA, 85.41 s for GraphicalVAR, and 518.16 s for GIMME. GraphicalVAR timed out on 1,271 of 21,600 fits. GIMME produced 4,846 valid outputs, 14,477 timeouts, and 2,277 post-estimation errors. DynEGA returned a valid output for every fit in the primary and extended designs. These runtimes are specific to our implementation and hardware. Appendix C gives the full runtime, factorial, and community-detection results.

Discussion

LOMM separates three determinants of multivariate time-series structure: the dynamics of latent processes, their expression over time, and their measurement by observed indicators. This separation matters when researchers use associations among classifier-score channels to infer dimensional structure. A change in the recovered grouping may reflect a different latent structure, weaker expression of an unchanged structure, or properties of the measurement process. LOMM allows these possibilities to be examined under controlled conditions by holding the generating dimensions and indicator assignments fixed while varying dynamics, activity, or measurement quality.

The two studies establish different parts of this contribution. Study 1 supports the statistical resemblance of calibrated bounded output to short MAFW classifier-score series. Study 2 establishes recovery performance for specified continuous-indicator models. The initialization and bounded-score analyses show why those recovery results require explicit conditions: changes in the starting states or observation transformation can substantially change the recovered structure.

Study 1 found that LOMM generated classifier-score series with high plausibility and broad coverage relative

to the MAFW reference. The static logistic-normal generator preserved the calibrated score geometry and between-video heterogeneity. It omitted temporal dependence and performed substantially worse on both measures. In this comparison, LOMM reproduced the joint statistical profile of the MAFW series more closely than the calibrated static logistic-normal generator. The metric-specific results also identify a concrete target for model development: LOMM captured most measures of level, dispersion, change, persistence, and cross-channel variation, while its entropy profile remained less similar to MAFW.

Study 2 evaluated recovery of a known factor structure from continuous indicators. DynEGA had the highest overall recovery in the primary matched-method design and was feasible at the scale of the extended factorial simulation, where it was the only method evaluated. Static EGA showed that most of the grouping was recoverable from contemporaneous levels; it exceeded GraphicalVAR in both dimensional recovery and assignment accuracy while running in a median of 2.15 s rather than 85.41 s. For this grouping target and these conditions, GraphicalVAR's additional lagged model did not improve recovery relative to static EGA. GraphicalVAR answers additional questions about lagged relations, which this comparison does not evaluate.

The trajectory diagnostics in Appendix C show that the ranking of methods depends on the recovery target. DynEGA's advantage concerned recovery of the number of dimensions and the indicator partition; alignment of recovered-group summaries with the latent trajectories among valid outputs with the correct number of dimensions produced a different ordering. The two targets should be evaluated separately. All comparisons apply to the specific estimation, network-conversion, and grouping procedures used here. GIMME's low overall recovery mainly reflected its failure to return valid outputs under a clip-level fitting procedure that differs from its usual use.

The most important qualification of the primary result concerns initialization. Recovery was lower with independent stationary starts for stable dimensions and bounded independent starts for amplifying dimensions than with the common fixed initialization. The primary initialization was therefore optimistic relative to this alternative. Because this sensitivity changed cross-dimension and cross-clip coherence together with the marginal initial-state distribution, it does not isolate the effect of sharing alone. Gradual activity transitions, small cross-loadings, and correlated acceleration inputs changed little. The bounded-score condition set a clearer limit: the softmax transformation changed the associations among the original indicators, and neither the raw scores nor the CLR components preserved the original grouping. The Study 2 recovery claims therefore apply to continuous indicators.

Limitations. Several limitations remain. The latent dimensions are uncoupled. The amplifying families produce large amplitudes late in the record and test recovery under those conditions only. The $T = 300$

conditions also change the observation interval, so they do not isolate the number of observations per clip. The GIMME residual-covariance conversion was constructed for this comparison and differs from typical idiographic use. Study 1 uses 100 short MAFW videos, one zero-shot scoring pipeline, and one LOMM specification chosen during model development on the same sample. LOMM is used here only as a generator with prespecified parameters; estimating it from data would require the identification constraints listed in Appendix A.

Three practical conclusions follow. Initial conditions can create structure unrelated to the intended measurement partition. Transforming indicators to bounded scores can change the structure among the analyzed variables. Computational feasibility belongs in any recovery evaluation, because a method that returns no output cannot recover the target. Applied researchers should therefore specify the latent dynamics, activity process, observation model, and recovery target before interpreting an empirical network. LOMM supports this workflow and can be extended to other scoring systems, coupled latent dimensions, and recovery targets defined directly for bounded data.

Declarations

Funding

The authors received no specific funding for this work.

Competing interests

The authors declare that they have no competing interests.

Ethics approval

This study used simulations and secondary video data obtained under the MAFW research-use terms. It involved no new recruitment, intervention, or direct collection of human-participant data by the authors.

Consent to participate

Not applicable.

Consent for publication

The manuscript includes four still frames from MAFW solely to illustrate the score-extraction procedure. The frames are cited to the dataset, and no source video is redistributed. Their inclusion follows the dataset's terms for limited reproduction in academic publications.

Availability of data and materials

The project GitHub repository at https://github.com/atomashevic/LOMM contains the simulation summary results, analysis-ready tables underlying the figures, and aggregate MAFW-derived summaries used in the simulated-versus-real comparison. Raw MAFW videos and frame-level classifier scores are not redistributed. Access to MAFW is governed by the dataset provider, and an end-to-end reanalysis of Study 1 requires approved access and local generation of the classifier scores.

Code availability

Code for generating LOMM datasets, fitting the methods, reproducing the statistical summaries, and regenerating the tables and figures is available in the same GitHub repository. The repository also provides the manuscript source and instructions for reproducing the analyses.

Authors' contributions

Aleksandar Tomašević contributed conceptualization, formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Hudson Golino contributed formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Alexander P. Christensen contributed methodology, software, writing the original draft, and reviewing and editing the manuscript.

Open Practices Statement

The study was not preregistered. The simulation and analysis code, manuscript source, paper-facing tables, and aggregate MAFW-derived summaries used in the simulated-versus-real comparison are available at https://github.com/atomashevic/LOMM. Raw MAFW videos are third-party materials, and the frame-level classifier scores are derived from those restricted materials; neither is included.

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Appendix A: Complete Specification of LOMM

This appendix gives the formal LOMM specification behind the conceptual account in the main text. It is intended for readers who want to reimplement the generator or examine its assumptions and identification limits.

A.1 The Latent Transition

The transition below advances the latent level and velocity over one observation interval while holding the acceleration input constant. The matrix $\Phi_f(\Delta)$ describes how the current state evolves, and $\Gamma_f(\Delta)$ describes how the input changes that state. For the oscillator in Equation 1, these are

$$\Phi_f(\Delta) = \exp(\mathbf{A}_f\Delta), \qquad \mathbf{A}_f = \begin{bmatrix} 0 & 1 \\ \eta_f & \zeta_f \end{bmatrix} \qquad \text{and} \qquad \Gamma_f(\Delta) = \int_0^\Delta \exp(\mathbf{A}_f u)\mathbf{b}\, du, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

For the nonzero position-feedback values used in the simulation grids, the input map has the equivalent exact form

$$\Gamma_f(\Delta) = \mathbf{A}_f^{-1}\{\Phi_f(\Delta) - \mathbf{I}_2\}\mathbf{b}.$$

The implemented state recurrence is

$$\begin{bmatrix} x_{if,n+1} \\ v_{if,n+1} \end{bmatrix} = \Phi_f(\Delta) \begin{bmatrix} x_{if,n} \\ v_{if,n} \end{bmatrix} + \Gamma_f(\Delta) q_{if,n}, \qquad n = 1, \ldots, T-1.$$

The integral definition remains valid when $\mathbf{A}_f$ is singular, including $\eta_f = 0$. If a root-based closed form is used numerically, a separate repeated-root branch is required when $\zeta_f^2 + 4\eta_f$ is close to zero.

The state components $x_{if,n}$ and $v_{if,n}$ are the level and its rate of change at t_n; η_f is position feedback, ζ_f is velocity feedback, and σ_{qf} is the standard deviation of the acceleration input. On the interval $[t_n, t_{n+1})$ the underlying continuous-time model is $\ddot{x}_{if}(t) = \eta_f x_{if}(t) + \zeta_f \dot{x}_{if}(t) + q_{if,n}$ with $q_{if,n}$ held constant, so $\Phi_f(\Delta)$ propagates the deterministic state and $\Gamma_f(\Delta)$ loads the input.

Let $\mathbf{q}_{i,n} = (q_{i1,n}, \ldots, q_{im,n})^\top$ and $\mathbf{D}_q = \mathrm{diag}(\sigma_{q1}, \ldots, \sigma_{qm})$. The generator draws

$$\mathbf{q}_{i,n} \sim \mathcal{N}_m\big(\mathbf{0}, \mathbf{D}_q \mathbf{R}_q \mathbf{D}_q\big).$$

The primary designs use $\mathbf{R}_q = \mathbf{I}_m$. The correlated-input sensitivity uses $\mathbf{R}_q = (1 - 0.15)\mathbf{I}_m + 0.15\mathbf{1}_m\mathbf{1}_m^\top$. Table B2 reports the Study 2 values of σ_{qf}.

In Study 1, σ_{qf} was set so that the stationary variance of $x_{if,n}$ was one. Specifically, if $\mathbf{P}_f^{(1)}$ solves $\mathbf{P}_f^{(1)} = \Phi_f \mathbf{P}_f^{(1)} \Phi_f^\top + \Gamma_f \Gamma_f^\top$, then $\sigma_{qf} = \{\mathbf{e}_1^\top \mathbf{P}_f^{(1)} \mathbf{e}_1\}^{-1/2}$.

Inputs are independent across clips and transitions and are independent of initial states, activity schedules, and measurement errors. They are also independent across dimensions when $\mathbf{R}_q = \mathbf{I}_m$. Each acceleration input is held constant for a full observation interval, a convention called zero-order hold. The input is therefore defined per interval; it is not a continuous-time white-noise process. The covariance added in one transition is $\Gamma_f(\Delta)\sigma_{qf}^2\Gamma_f(\Delta)^\top$ for a single dimension.

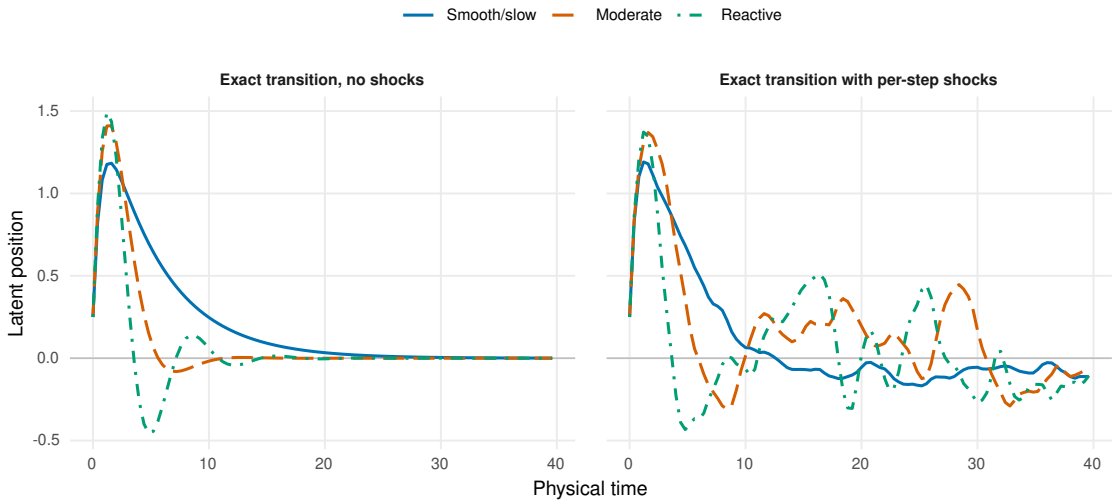


Figure A1: Illustrative latent trajectories for three uncoupled dimensions under the exact transition, shown without and with piecewise-constant acceleration inputs. The inputs create irregular peaks and reversals while the feedback parameters govern the underlying dynamic regime.

A.2 Dynamic Regimes and Finite-Horizon Amplification

The characteristic roots describe whether the deterministic motion decays, grows, or oscillates. For dimension f, they are

$$r_{f,\pm} = \frac{\zeta_f \pm \sqrt{\zeta_f^2 + 4\eta_f}}{2}.$$

Asymptotic stability means that the effect of an initial state eventually decays to zero in the absence of new inputs. In continuous time, it requires $\eta_f < 0$ and $\zeta_f < 0$, and the roots are oscillatory when $\zeta_f^2 + 4\eta_f < 0$. For

the sampled transition, the spectral radius ρ is the largest absolute eigenvalue of $\Phi_f(\Delta)$. Stability requires

$$\rho\{\Phi_f(\Delta)\} = \exp\big[\Delta \max\{\mathrm{Re}(r_{f,+}), \mathrm{Re}(r_{f,-})\}\big] < 1.$$

The real parts of the characteristic roots determine whether the effect of an initial state decays or grows. Over the observation horizon $H = (T-1)\Delta$, we summarize this modal change by

$$g_f(H) = \exp\big[H \max\{\mathrm{Re}(r_{f,+}), \mathrm{Re}(r_{f,-})\}\big].$$

This quantity describes deterministic modal growth or decay. It does not bound individual stochastic trajectories or capture all possible transient amplification. Table C6 reports the maximum real part of the roots and the modal factors, while Table C5 summarizes the amplitudes actually produced by the generator. Activity modulation can create additional nonstationarity and is reported separately. The amplifying configurations are finite-record stress tests.

A.3 Initial States

The system is uncoupled: η_f, ζ_f, and σ_{qf} may vary by dimension, and a dimension does not directly change the level or velocity of another dimension.

Study 1 initialized every dimension independently at the start of each uninterrupted segment by drawing $\mathbf{s}_{if,1} = (x_{if,1}, v_{if,1})^\top$ from its zero-mean stationary Gaussian distribution. A new state was drawn after each interruption.

Study 2's primary simulation design initialized every dimension in every clip at $x_{if,1} = 0.25$ and $v_{if,1} = 2$. The sensitivity analysis replaced these common fixed values with draws that were independent across dimensions and clips.

For a stable dimension, the stationary covariance $\mathbf{P}_f$ solves

$$\mathbf{P}_f = \Phi_f(\Delta)\mathbf{P}_f\Phi_f(\Delta)^\top + \Gamma_f(\Delta)\sigma_{qf}^2\Gamma_f(\Delta)^\top,$$

whose solution is unique when $\rho\{\Phi_f(\Delta)\} < 1$. Drawing $\mathbf{s}_{if,1} \sim \mathcal{N}(\mathbf{0}, \mathbf{P}_f)$ makes the state process stationary before the activity layer is applied; time-varying activity modulation still renders $a_{if}(t)x_{if}(t)$ nonstationary.

For a dimension with positive velocity feedback, the sensitivity draws $x_{if,1} \sim \mathrm{Unif}(-0.25, 0.25)$ and $v_{if,1} \sim$

Unif$(-2, 2)$. The alternative therefore changes cross-dimension and cross-clip coherence, marginal means, covariance, stationarity, and the magnitude and direction of the initial transient; the observed horizon is unchanged.

A.4 The Activity Layer at a Transition

Changes in an observed indicator can arise from changes in the latent level or changes in its activity multiplier. For the expressed signal $\tilde{x}_{if,n} = a_{if,n} x_{if,n}$,

$$\frac{\tilde{x}_{if,n+1} - \tilde{x}_{if,n}}{\Delta} = a_{if,n+1} \frac{x_{if,n+1} - x_{if,n}}{\Delta} + \frac{a_{if,n+1} - a_{if,n}}{\Delta} x_{if,n}.$$

The first term is latent change expressed at the new activity level. The second is change caused by the activity multiplier. An abrupt activity transition can therefore contribute strongly to an estimated derivative even though the latent level remains continuous. The gradual-transition sensitivity analysis examines whether recovery depends on these abrupt changes. At each observation, activity multiplies the completed latent state at the same observation index.

The activity schedule takes two nonnegative values, a_{high} and a_{low}. In non-switching families the dominant dimension remains at a_{high} for the whole record and the others remain at a_{low}. In switching families, dwell fractions determine how long each dimension remains dominant before the schedule reassigns a_{high}. Appendix B lists the numeric gains and dwell values used in the Study 2 families.

A.5 The Observation Layer

LOMM has two observation models. The continuous-indicator model maps latent processes directly to observed variables with specified loadings and measurement error. The bounded-score model first generates log-ratio coordinates and then converts them to positive scores that sum to one. Its ILR coordinates combine the logarithms of multiple score channels; there is no separate ILR coordinate for each channel. We use separate notation for continuous indicators, $\mathbf{z}$, and log-ratio coordinates, $\mathbf{u}$.

Let J denote the number of positive score channels, let $d = J - 1$, and let $\mathbf{V} \in \mathbb{R}^{J \times d}$ be a fixed orthonormal isometric log-ratio (ILR) basis satisfying $\mathbf{V}^\top \mathbf{V} = \mathbf{I}_d$ and $\mathbf{V}^\top \mathbf{1}_J = \mathbf{0}$. The Study 1 basis used here is a normalized Helmert basis constructed in the stated 11-channel order. The full observation model in log-ratio coordinates is

$$\mathbf{u}_{in} = \mu^{(u)} + \mathbf{b}_i^{(u)} + \kappa s_i^{(u)} \left(\Lambda_i^{(u)} \tilde{\mathbf{x}}_{in} + \varepsilon_{in}^{(u)} \right).$$

Here $\mathbf{u}_{in}, \mu^{(u)}, \mathbf{b}_i^{(u)}, \varepsilon_{in}^{(u)} \in \mathbb{R}^d$, $s_i^{(u)} > 0$ is the clip scale, $\kappa > 0$ is a global scale, and $\Lambda_i^{(u)} \in \mathbb{R}^{d \times m}$ maps the m activity-scaled latent positions into the log-ratio coordinates. The residual covariance is $\Sigma_i^{(u)} \in \mathbb{R}^{d \times d}$, with $\varepsilon_{in}^{(u)} \mid \Sigma_i^{(u)} \sim \mathcal{N}(\mathbf{0}, \Sigma_i^{(u)})$. The clip scale multiplies both signal and residual, and the baseline is additive.

The continuous-indicator branch has output dimension p and uses separate measurement parameters:

$$\mathbf{z}_{in} = \mathbf{b}_i^{(z)} + s_i^{(z)} \left(\Lambda_i^{(z)} \tilde{\mathbf{x}}_{in} + \varepsilon_{in}^{(z)} \right), \qquad \varepsilon_{in}^{(z)} \mid \Psi_i \sim \mathcal{N}(\mathbf{0}, \Psi_i),$$

where $\mathbf{z}_{in}, \mathbf{b}_i^{(z)}, \varepsilon_{in}^{(z)} \in \mathbb{R}^p$, $\Lambda_i^{(z)} \in \mathbb{R}^{p \times m}$, $\Psi_i \in \mathbb{R}^{p \times p}$, and $s_i^{(z)} > 0$. In both branches, residuals are conditionally independent over time and across clips and are independent of latent states, initial states, activity schedules, and acceleration inputs. Designs with fixed measurement parameters use a common loading matrix and residual covariance within the relevant branch.

The residual-correlation sensitivity condition applies to the continuous indicators and sets

$$\Psi = s_\varepsilon^2 \left\{ (1 - \rho) \mathbf{I}_p + \rho \mathbf{1}_p \mathbf{1}_p^\top \right\}, \qquad \rho = 0.10,$$

which introduces residual covariance between every pair of the p indicators, including pairs from different prespecified blocks.

The baseline constructions are also branch-specific. Study 1 draws $\mathbf{b}_i^{(u)}$ from a zero-mean Gaussian whose covariance is estimated from MAFW calibration videos in the full d-dimensional log-ratio coordinate space. Study 2's between-clip-variability analysis uses

$$\mathbf{b}_i^{(z)} = \Lambda^{(z)} \beta_i, \qquad E(\beta_i) = \mathbf{0},$$

so that between-clip variation stays within the column space of the continuous-indicator loading matrix. Study 2 conditions without baseline and scale differences set $\mathbf{b}_i^{(z)} = \mathbf{0}$ and $s_i^{(z)} = 1$, giving

$$\mathbf{z}_{in} = \Lambda_i^{(z)} \tilde{\mathbf{x}}_{in} + \varepsilon_{in}^{(z)}.$$

This equation defines the continuous indicators used in Study 2's primary recovery designs before any bounded-score mapping. Each of the $p = mk$ indicators loads on one dimension and has diagonal measurement error. In the matched-method and extended factorial designs, each primary loading receives an independent Unif$(-0.15, 0.15)$ deviation from its nominal value, and each residual variance receives an independent Unif$(-0.01, 0.01)$ deviation from s_ε^2. The one-at-a-time sensitivity and between-clip-variability designs use fixed equal values except where the stated condition changes them. Zero baseline and unit scale remove baseline and overall-scale differences; common loadings and residual covariance require the separate fixed-parameter condition.

For bounded output, LOMM converts the coordinates to strictly positive scores that sum to one, the open probability simplex. The centered log-ratio (CLR) expresses each channel relative to the geometric mean of all channels. The isometric log-ratio (ILR) represents the same information in $d = J - 1$ orthonormal coordinates. Formally,

$$\mathrm{clr}(\mathbf{p}) = \log \mathbf{p} - J^{-1}\{\mathbf{1}_J^\top \log(\mathbf{p})\}\mathbf{1}_J, \qquad \mathrm{ilr}_{\mathbf{V}}(\mathbf{p}) = \mathbf{V}^\top \mathrm{clr}(\mathbf{p}),$$

with the logarithm applied componentwise. For the canonical bounded LOMM output, applying softmax to $\mathbf{V}\mathbf{u}_{in}/\tau$ gives

$$\mathrm{ilr}_{\mathbf{V}}(\mathbf{p}_{in}) = \frac{\mathbf{u}_{in}}{\tau}, \qquad \mathrm{clr}(\mathbf{p}_{in}) = \frac{\mathbf{V}\mathbf{u}_{in}}{\tau},$$

The second identity follows because the components of $\mathbf{V}\mathbf{u}_{in}$ sum to zero. The fixed basis excludes the unidentified common-logit direction and the softmax produces strictly positive rows summing to one, so the scores identify the scaled coordinates $\mathbf{u}_{in}/\tau$ exactly. A common-logit component remains unobservable, and the unscaled coordinates and τ are not separately identified without an external scale constraint.

The Study 2 bounded-score sensitivity condition uses the $p = mk$ continuous-indicator vector directly. With $J = p$, it computes

$$\mathbf{p}_{in}^{\mathrm{dep}} = \mathrm{softmax}\left(\frac{\mathbf{z}_{in}}{\tau}\right), \qquad \mathrm{clr}\left(\mathbf{p}_{in}^{\mathrm{dep}}\right) = \frac{1}{\tau}\mathbf{P}_p\mathbf{z}_{in}, \qquad \mathbf{P}_p = \mathbf{I}_p - \frac{\mathbf{1}_p\mathbf{1}_p^\top}{p}.$$

For any orthonormal ILR basis $\mathbf{V}_p \in \mathbb{R}^{p\times(p-1)}$, $\mathbf{V}_p\mathbf{V}_p^\top = \mathbf{P}_p$. Defining $\mathbf{u}_{in}^{\mathrm{dep}} = \mathbf{V}_p^\top\mathbf{z}_{in}$ therefore gives $\mathbf{V}_p\mathbf{u}_{in}^{\mathrm{dep}} = \mathbf{P}_p\mathbf{z}_{in}$. Softmax is invariant to the removed row mean, so the direct-softmax branch is equivalent to applying the basis form to $\mathbf{u}_{in}^{\mathrm{dep}}$. The analysis used the p raw scores and the p CLR components, which

satisfy a zero-sum constraint; it did not analyze the $(p-1)$ ILR coordinates. The common-logit direction in $\mathbf{z}_{in}$ is unidentifiable after this transformation.

In the equal-loading reference condition used for this sensitivity, an equal increase in all expressed latent levels adds the same amount to every logit. Softmax removes this common component. Specifically, $\Lambda^{(z)}\mathbf{1}_m = \lambda\mathbf{1}_p$, so $\mathbf{P}_p\Lambda^{(z)}\mathbf{1}_m = \mathbf{0}$. The projected loading matrix $\mathbf{P}_p\Lambda^{(z)}$ has rank $m-1$, leaving one fewer independent latent direction. The transformation therefore changes the latent representation as well as the associations among indicators. The original channel grouping may still contain recoverable information; this sensitivity analysis tests whether methods recover that grouping after the transformation.

A.6 Converting Period and Damping to Feedback Parameters

The Study 1 calibration search specified dynamics by natural period and damping ratio. The following equations convert those settings to the feedback parameters η_f and ζ_f. If P_f is the natural period in observation intervals and ξ_f is the damping ratio, then

$$\omega_{n,f} = \frac{2\pi}{P_f\Delta}, \qquad \eta_f = -\omega_{n,f}^2, \qquad \zeta_f = -2\xi_f\omega_{n,f},$$

and the damped angular frequency is $\omega_{d,f} = \omega_{n,f}\sqrt{1-\xi_f^2}$. With the Study 1 settings $P_f = 12$, $\xi_f = 0.20$, and $\Delta = 1$, these give $\eta_f = -0.2742$ and $\zeta_f = -0.2094$. The acceleration inputs were scaled to give unit stationary position variance.

A.7 Identifiability

The simulations fix the generating parameters in advance. Estimating them from observed data would require additional constraints because different parameter combinations can produce the same observations. The main ambiguities concern scale, dimension labels, and sampling frequency.

For either observation branch $h \in \{u, z\}$ and any nonsingular diagonal matrix $\mathbf{D}$,

$$\Lambda^{(h)}\tilde{\mathbf{x}}_{in} = (\Lambda^{(h)}\mathbf{D}^{-1})(\mathbf{D}\tilde{\mathbf{x}}_{in}),$$

so loading magnitude and latent-trajectory scale are not jointly identified without an explicit scale constraint. Activity gain and acceleration-input scale contribute to latent amplitude in the same way. The softmax layer adds invariance to a common location shift and a tradeoff between the temperature and the coordinate scale.

Dimension labels are recoverable only up to permutation.

Sampling creates another ambiguity: different oscillation frequencies can produce the same values at the observation times. This is called aliasing. For an underdamped dimension, write

$$r_{f,\pm} = \frac{\zeta_f}{2} \pm i\omega_f, \qquad \omega_f = \frac{1}{2}\sqrt{-(\zeta_f^2 + 4\eta_f)}.$$

At observation times $t_n = (n-1)\Delta$, the frequencies ω_f and $\omega_f + 2\pi q/\Delta$ for integer q generate the same sampled sine and cosine sequences after an appropriate change in initial velocity. Identifying continuous-time oscillator parameters from data therefore requires a cadence-dependent restriction such as

$$0 < \omega_f < \frac{\pi}{\Delta},$$

together with a specified acceleration-input model and constraints on the latent state basis. Together, these ambiguities require sign and scale constraints on the dimensions, normalization of the activity schedule, anchoring of either the loading scale or the input scale, a reference-logit or sum-to-zero constraint, a fixed or otherwise identified temperature, and this frequency restriction.

Every oscillatory mode used here satisfies the frequency restriction. In Study 1, the selected damped angular frequency is approximately 0.513 radians per observation; the largest frequency in the calibration search is approximately 0.524. Both are below the sampling bound of π. Study 2 uses $\Delta \in \{0.25, 0.4\}$, giving bounds of approximately 12.57 and 7.85 against a largest angular frequency of approximately 1.04. The simulations do not estimate any of these parameters.

Appendix B: Study Designs and Implementation Details

This appendix gives the design and implementation details for both studies. Study 1 evaluates resemblance to real bounded-score series. Study 2 evaluates recovery of a prespecified factor structure from continuous indicators.

B.1 Study 1 Scoring and Calibration

The sample contains 100 MAFW videos selected across the available expression categories. Category labels served only to stratify sampling and fold construction. The analysis retained 3,557 complete observations in 188 uninterrupted segments. Individual videos contain 16 to 50 complete observations, with a median of 35. Segment lengths range from 1 to 50 observations, with a median of 13.5.

transforEmotion sampled up to 50 frames uniformly from each video and used OpenAI CLIP ViT-Large/14. A Haar cascade detected faces, and the largest detected face was expanded by 50 pixels on each side before classification. Frames without a detected face received missing scores. Eleven lowercase labels were passed directly to the text encoder: anger, disgust, fear, happiness, neutral, sadness, surprise, contempt, anxiety, helplessness, and disappointment. Burned-in subtitles were not masked when they fell inside the padded face crop.

Scores were required to lie in $[0, 1]$. A complete row was retained when its sum was positive, finite, and within 10^{-6} of one, then divided by that sum in the fixed channel order. Values were not truncated, and zero replacement was disabled. The transformation stops with an error if a complete row contains a zero. None of the 39,127 retained scores equaled zero; the minimum before row normalization was approximately 2.42×10^{-5}. The analysis used complete observations and preserved the uninterrupted segment boundaries.

Features and segment boundaries. The evaluation panel contains eleven statistics: median level, interquartile range of levels, interquartile range of changes, lag-1 autocorrelation, run-length entropy, sample entropy, permutation entropy, spectral entropy, effective rank, robust level variance, and robust change variance. The eight univariate statistics were computed for each of the eleven score channels and each of their eleven CLR coordinates. The three joint statistics used the score panel and its ten ILR coordinates. For each video, the median of the finite channel values summarized each univariate statistic. Together with the joint statistics, these summaries give 22 features. A feature remained undefined if no channel had a finite value.

Levels were pooled across all retained observations within a video. Changes and lagged pairs were formed within segments and then pooled across the video; no pair crossed a segment boundary. Lag-1 correlation was computed from these pooled pairs. Run lengths were formed separately within segments, after removing

zero changes, capped at eight, and pooled across segments. Sample and permutation entropy pooled windows formed entirely within segments. Sample entropy compared these valid windows across the video, including windows from different segments. Spectral entropy was computed separately for each eligible segment and averaged using segment length minus one as the weight. A segment containing one observation contributed to level statistics but supplied no changes, lagged pairs, or entropy windows.

Change IQR and robust change variance required at least one within-segment change. Lag-1 correlation required at least two pairs and nonzero variation in the first and second observations in the pairs. Run-length entropy required four observations across the video and was zero when fewer than two runs remained.

Sample entropy used order 2, windows of three observations, and a tolerance of 0.20 times the channel's standard deviation across the video. It required at least two windows and used the negative logarithm of $(A+0.5)/(B+1)$, where A counts matching pairs of three-observation windows and B counts matches between their first two observations; a zero or undefined channel standard deviation produced zero. Permutation entropy used order 3, required at least two windows, and resolved ties by position within the window.

Spectral entropy required segments of at least eight observations with positive total spectral power and at least two positive frequency powers. Statistics failing their requirements remained undefined.

Effective rank used the positive eigenvalues of the video correlation matrix and required that matrix to be finite. Robust level variance was the 10% trimmed mean squared distance from the video center divided by the coordinate dimension. Robust change variance was the 10% trimmed mean squared within-segment first difference divided by twice the coordinate dimension. The joint statistics required at least two observations.

Calibration. The 100 videos were divided into four category-stratified folds. Each fold used 75 calibration videos and 25 evaluation videos. The calibration videos determined the coordinate center, loading subspace, between-video baseline covariance, and empirical distribution of within-video scale. The first three eigenvectors of the pooled within-video covariance supplied the loading directions. The between-video baseline covariance was shrunk halfway toward its diagonal, and its eigenvalues were floored at 10^{-8}. The generator included between-video baseline and scale variation. Activity modulation was disabled: every latent dimension had an activity multiplier of one at every generated observation.

Generator settings were selected within each calibration set. The dynamic search crossed periods of 12, 24, and 48 observations with damping ratios of 0.01, 0.05, and 0.20. Selection used the interquartile range of changes, lag-1 autocorrelation, run-length entropy, and spectral entropy. Conditional on the selected dynamics, the measurement search crossed residual standard deviations of 0.10, 0.20, and 0.30 with κ values of 0.75, 1.00, and 1.25. Measurement selection used 16 discrepancy components: median level, level IQR,

change IQR, lag-1 autocorrelation, run-length entropy, spectral entropy, effective rank, and total variance, each for scores and log-ratios. Total variance was the sum of the coordinate variances. The additional entropy and robust variance statistics belonged to the subsequent 22-feature evaluation. Each candidate was evaluated with three calibration simulations. All four folds selected a period of 12 observations, damping ratio 0.20, residual standard deviation 0.30, and $\kappa = 0.75$.

The static logistic-normal generator used the same calibrated coordinate center, covariance geometry, between-video baseline and scale distributions, observation positions, and segment boundaries. It drew independent log-ratio observations within each video. Each selected generator produced 100 datasets per fold with 25 simulated series in each dataset.

Evaluation. The MAFW reference used 1,000 comparisons per fold. Each comparison drew two disjoint, category-stratified groups of 25 videos from the 75 calibration videos. Every feature was standardized by the median and scale of its finite values in the calibration videos. The scale was the IQR, with successive fallbacks to 1.4826 times the median absolute deviation, the standard deviation, and one; a data-derived scale had to exceed 10^{-8}. Before standardization, any undefined video feature was replaced by its calibration median and therefore became zero after standardization. This replacement used no evaluation videos. One MAFW video had no segment of eight observations, leaving its two spectral-entropy features undefined; these were the only two undefined values among the 2,200 empirical video features. The distance between two series was the root-mean-square Euclidean distance across all 22 standardized features, including features replaced in this way. The fold-specific similarity threshold was the 95th percentile of the pooled directional nearest-neighbor distances between the two MAFW groups.

For each generated dataset, plausibility was the proportion of its 25 simulated series whose nearest MAFW evaluation series fell within the threshold. Coverage was the proportion of the 25 MAFW evaluation series whose nearest simulated series fell within the threshold. Results were averaged across the four folds. The reported ranges are the 2.5th and 97.5th percentiles across the 100 generated datasets or 1,000 MAFW reference comparisons.

Distributional comparisons. Calibration and metric-specific distributional discrepancy used finite channel-by-video contributions for each univariate statistic, and one contribution per video for each joint statistic. For univariate statistics, these distributions pooled channel values across videos. Undefined contributions were omitted separately from the empirical and simulated distributions. Their absolute median difference and one-dimensional Wasserstein distance were divided by the calibration scale, using the same scale fallback rule, and the larger standardized value was retained. The measurement objective averaged its 16 components,

giving equal weight to scores and log-ratios; an entirely undefined component made the full selection loss infinite. All stored calibration objectives were finite. The reported full distributional discrepancy averaged the 22 evaluation components.

B.2 Study 2 Simulation Designs

For each clip, LOMM sets or draws the starting state and acceleration inputs according to the configuration, advances each latent dimension, applies the activity schedule, and constructs the observed indicators. The primary, shorter-series, and extended factorial designs use continuous indicators. All clips share the same indicator assignments and nominal measurement parameters. The generator draws clip-specific loading and residual-variance perturbations as specified in the random-draw table below. The one-at-a-time sensitivity conditions use fixed equal measurement parameters except for the stated change.

Table B1: The six Study 2 scenario families. Dwell values are fractions of the number of observations per clip.

Code	Family	Switching	Active / background gain	Dwell fraction	Dynamic rule
A	Weak expression, one dominant dimension	No	0.35 / 0.05	1.00 / 1.00	Stable, low input scale
B	Strong expression, one dominant dimension	No	1.00 / 0.08	1.00 / 1.00	Stable, standard input scale
C	Strong expression, switching dominant dimension	Yes	1.00 / 0.08	0.20 / 0.45	Stable, standard input scale
D	Strong expression, rapid switching	Yes	1.25 / 0.08	0.03 / 0.10	Stable, high input scale
E	Amplifying dynamics, one dominant dimension	No	1.00 / 0.08	1.00 / 1.00	Positive velocity feedback
F	Amplifying dynamics, switching dominant dimension	Yes	1.10 / 0.10	0.12 / 0.30	Positive velocity feedback

In non-switching scenarios, one dimension remains dominant for a whole clip and the dominant dimension rotates across clips. Switching scenarios sample dwell intervals and change the dominant dimension over time. Families A through D use stable dynamics. Families E and F use positive velocity feedback and test recovery under finite-horizon amplification.

The dynamic parameters are listed in Table B2. Each vector gives one value per latent dimension. The primary and shorter designs use $\Delta = 0.4$; the extended design with 300 observations uses $\Delta = 0.25$. The displayed σ_q values are input standard deviations per transition. Holding them fixed while changing Δ changes the input specification per unit of time.

Table B2: Dynamic parameters for each scenario family. Entries in each vector follow the dimension order. Acceleration-input SD is the per-step standard deviation σ_q. Four-dimensional conditions occur only in the extended factorial simulation.

Code	m	η	ζ	Acceleration-input SD
A	2D	-0.30; -0.55	-1.70; -1.00	0.06; 0.15
A	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.06; 0.15; 0.23
A	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.06; 0.15; 0.23; 0.33
B	2D	-0.30; -0.55	-1.70; -1.00	0.12; 0.30
B	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.12; 0.30; 0.45
B	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.12; 0.30; 0.45; 0.65
C	2D	-0.30; -0.55	-1.70; -1.00	0.12; 0.30
C	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.12; 0.30; 0.45
C	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.12; 0.30; 0.45; 0.65
D	2D	-0.30; -0.55	-1.70; -1.00	0.15; 0.38
D	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.15; 0.38; 0.56
D	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.15; 0.38; 0.56; 0.81
E	2D	-0.05; -0.10	0.08; 0.10	0.25; 0.25
E	3D	-0.05; -0.10; -0.15	0.08; 0.10; 0.12	0.25; 0.25; 0.25
E	4D	-0.05; -0.10; -0.15; -0.20	0.08; 0.10; 0.12; 0.14	0.25; 0.25; 0.25; 0.25
F	2D	-0.05; -0.10	0.08; 0.10	0.35; 0.35
F	3D	-0.05; -0.10; -0.15	0.08; 0.10; 0.12	0.35; 0.35; 0.35
F	4D	-0.05; -0.10; -0.15; -0.20	0.08; 0.10; 0.12; 0.14	0.35; 0.35; 0.35; 0.35

The primary $T = 100$ design contains 21,600 datasets: 216 factorial conditions with 100 paired replications per condition. The factorial crosses six scenario families, $m \in \{2, 3\}$ dimensions, $N \in \{50, 100\}$ clips, loadings $\{0.40, 0.60, 0.80\}$, residual standard deviations $\{0.125, 0.25, 0.50\}$, and four indicators per dimension. The shorter-series design generates 21,600 corresponding datasets with $T = 25$, $\Delta = 0.4$, and duration 9.6. Every method receives the same dataset within each condition and replication.

The one-at-a-time sensitivity analyses fix $N = 100$, $T = 100$, loading 0.60, residual standard deviation 0.25, and four indicators per dimension (the reference condition). They cross all six families, two dimensionalities, and 50 paired replications while changing one feature at a time.

Table B3: One-at-a-time sensitivity conditions. Each changes one design component of the reference condition.

Sensitivity condition	Specification
Alternative initialization	Stationary zero-mean starts for stable dimensions; bounded independent starts for amplifying dimensions
Gradual activity transitions	A five-observation transition replaces each abrupt switch
Cross-loadings	Normal cross-loadings with pre-truncation SD 0.05, truncated to [-0.10, 0.10]
Residual equicorrelation	Correlation 0.10 between every residual pair, including pairs from different dimensions
Correlated inputs	Acceleration-input correlation 0.15 between dimensions
Bounded scores	Direct softmax of the p indicators with tau = 1, analyzed as raw scores or CLR components

Table B4: Random draws and perturbations.

Component	Scope	Specification
Acceleration input	Studies 1 and 2	$\mathbf{q}_{i,n} \sim \mathcal{N}_m(\mathbf{0}, \mathbf{D}_q \mathbf{R}_q \mathbf{D}_q)$. The primary designs use $\mathbf{R}_q = \mathbf{I}_m$; the correlated-input sensitivity sets every off-diagonal correlation to 0.15. Draws are independent across clips and transitions. Study 1 scales σ_{qf} to give unit stationary position variance; Table B2 gives the Study 2 values.
Initial state	Study 1	At the start of each uninterrupted segment, each dimension is drawn independently as $\mathbf{s}_{if,1} \sim \mathcal{N}_2(\mathbf{0}, \mathbf{P}_f)$, where $\mathbf{P}_f$ is the stationary covariance for the selected transition and input scale.
Activity	Study 1	Activity modulation is disabled: $a_{if,n} = 1$ for every dimension and generated observation.
Primary loadings	Study 2 matched and extended designs	For item j assigned to dimension f, $\Lambda^{(z)}_{i,jf} = \lambda + U_{ij}$ with independent $U_{ij} \sim \text{Unif}(-0.15, 0.15)$. All cross-loadings are zero.
Residual variances	Study 2 matched and extended designs	For indicator j, $\Psi_{i,jj} = s^2_\varepsilon + V_{ij}$ with independent $V_{ij} \sim \text{Unif}(-0.01, 0.01)$; off-diagonal entries are zero.
Switching dwell	Study 2 switching families	Set $L_{\min} = \max\{1, \lfloor T d_{\min} \rfloor\}$ and $L_{\max} = \max\{L_{\min}, \lceil T d_{\max} \rceil\}$. Each dwell is sampled uniformly from the integers $L_{\min}, \ldots, L_{\max}$; the final dwell is truncated at observation T. The first dominant dimension rotates across clips, and each later dominant dimension is sampled uniformly from the other $m-1$ dimensions.
Cross-loadings	Study 2 sensitivity	Each non-primary loading is drawn independently from $\mathcal{N}(0, 0.05^2)$ and truncated componentwise to $[-0.10, 0.10]$. Primary loadings remain fixed at 0.60.
Factor baseline β_i	Study 2 between-clip analysis	$\beta_i \sim \mathcal{N}_m(\mathbf{0}, 1.8487025519^2 \mathbf{I}_m)$ and $\mathbf{b}^{(z)}_i = \Lambda^{(z)} \beta_i$.
Clip scale $s^{(z)}_i$	Study 2 between-clip analysis	$s^{(z)}_i \sim \text{Unif}(0.5, 1.5)$, independently across clips. It multiplies the within-clip signal and measurement residual before the baseline is added.

The between-clip analysis uses 80 paired datasets from stable families A through D, $m \in \{2, 3\}$, and 10 replications. Each dataset has $N = 100$, $T = 100$, four indicators per dimension, loading 0.60, residual standard deviation 0.25, and independent stationary starts. Its baseline varies within the loading space according to $\beta_i \sim \mathcal{N}_m(\mathbf{0}, 1.8487025519^2 \mathbf{I}_m)$, and its clip scale follows $s^{(z)}_i \sim \text{Unif}(0.5, 1.5)$.

The extended factorial simulation evaluates DynEGA on 518,400 datasets. It crosses $N \in \{50, 100\}$, $T \in \{100, 300\}$, $m \in \{2, 3, 4\}$, $k \in \{4, 6\}$, three loading levels, three residual-error levels, six scenarios, and 400 replications. The $T = 100$ conditions use $\Delta = 0.4$ and the $T = 300$ conditions use $\Delta = 0.25$ while holding per-step input variance fixed. Their difference therefore combines the number of observations, the observation interval, the physical duration, and the discrete input specification.

The public lomm_observe() function implements the observation step and records the configuration, seeds, and hashes needed for reproduction. Study 2 uses the benchmark generator documented in the analysis code, which returns continuous indicators for the primary designs and applies the direct-softmax and CLR transformation for the bounded-score sensitivity condition.

B.3 Method Implementations

DynEGA estimates first derivatives from time-delay embeddings, constructs an EBICglasso partial-correlation network, and groups its positive edges (Boker et al., 2010; H. Golino et al., 2020; H. Golino et al., 2022; H. F. Golino & Epskamp, 2017). Video identity prevents embedding windows from crossing clip boundaries. An explicit search selects the embedding dimension by the Total Entropy Fit Index, with shorter embeddings breaking ties. Candidate dimensions are $\{5, 10, 15, 25, 50\}$ for $T = 100$ and $\{5, 6, 10, 12\}$ for $T = 25$. EGAnet 2.3.0 uses GLLA embedding parameters tau = 1 and delta = 1, first derivatives, automatic correlations, pairwise missing-data handling, and the glasso model. The derivative scale does not affect the correlation matrix because conversion from observation units to physical time multiplies every derivative by the same positive constant.

Static EGA applies EBICglasso to the contemporaneous correlation matrix of the stacked indicator levels and then uses the same network-processing and community-detection rules as DynEGA. It ignores the observation order. This makes it a benchmark that shows how much of the grouping is recoverable without temporal information, and a computational benchmark for the temporal procedures.

GraphicalVAR uses mlGraphicalVAR with clip identity as the grouping variable, within-clip centering, variable standardization, and gamma 0.5 (Epskamp et al., 2018; Epskamp, 2024). Lagged pairs never cross clip boundaries. The grouping uses the returned fixedPCC contemporaneous residual partial-correlation matrix. The diagonal is removed and the matrix is treated as an undirected weighted graph.

GIMME fits each standardized clip separately with unified structural equation modeling and vector autoregression (Gates & Molenaar, 2012; Lane et al., 2019). Standardized residual covariance matrices from successful fits are aligned by variable name, averaged across clips, and symmetrized. The resulting graph is a residual-covariance graph. Unequal residual variances can therefore influence its weights. This conversion was built for the present comparison and differs from typical idiographic use.

For every method, negative graph weights are set to zero. If no positive edge remains, absolute values of the original off-diagonal weights are used; only an all-zero matrix produces singleton communities. The integrated outputs do not record the frequency of this fallback. Louvain at resolution 1.0 is primary (Blondel et al., 2008). Leiden and Walktrap are applied to the same fitted network as sensitivity analyses (Pons & Latapy, 2005; Traag et al., 2019). A deterministic offset from the dataset seed initializes each method fit.

GraphicalVAR and GIMME were omitted from the 518,400-dataset extended factorial simulation because the full design was computationally impractical. Both were included in the primary and shorter-series designs. GraphicalVAR was also included in the one-at-a-time sensitivity and between-clip analyses. GIMME was

included in the between-clip analysis and excluded from the one-at-a-time sensitivity analyses.

B.4 Outcomes, Uncertainty, and Computing

A method can fail to recover the generating dimensions because it returns no usable network or because its network yields the wrong number of groups. The primary recovery rate counts both outcomes as failures and uses all attempted datasets as its denominator. We also report the proportion of attempts that return a valid network and the proportion of valid outputs with the correct number of dimensions. These supporting rates separate computational failure from incorrect grouping.

Adjusted Rand index and variation of information compare the full estimated and prespecified partitions among valid outputs. Higher adjusted Rand index and lower variation of information indicate closer agreement. Assignment accuracy matches estimated groups to dimensions to maximize correct indicator assignments. Every indicator remains in the denominator, and indicators in unmatched groups count as incorrect.

Trajectory diagnostics assess whether the recovered groups summarize the time course of the generating processes. They are constructed from the levels of the analyzed indicators after grouping, including for DynEGA. Scores retain the original observation order within each clip. The indicators are standardized over the pooled clips, and each recovered group is summarized by its first principal component. A single-indicator group uses its standardized indicator, and a row mean is used if a component cannot be computed. Each resulting summary is then standardized over the pooled clips. Let $\mathbf{y}_{ig}$ denote the summary of recovered group g in clip i, and let $\tilde{\mathbf{x}}_{if}$ denote the activity-scaled trajectory of data-generating dimension f. The best-match absolute trajectory correlation is

$$R_{\text{best}} = \frac{1}{Nm} \sum_{i=1}^{N} \sum_{f=1}^{m} \max_{g} \left| \text{cor}\left(\tilde{\mathbf{x}}_{if}, \mathbf{y}_{ig}\right) \right|.$$

Correlations use pairwise complete observations. The maximizing recovered group is selected separately for each data-generating dimension, so the same recovered-group summary may be selected for multiple dimensions.

Normalized root mean squared error (NRMSE) measures the difference between the standardized trajectories. It uses a separate one-to-one assignment within each clip. For data-generating dimension f and recovered group g, the sign-invariant cost is

$$d_{ifg} = \min_{\delta \in \{-1,1\}} \left[\frac{1}{K_{ifg}} \sum_{n \in \mathcal{O}_{ifg}} \left\{ \frac{\tilde{x}_{if,n} - \bar{x}_{if}}{s_{x,if}} - \delta \frac{y_{ig,n} - \bar{y}_{ig}}{s_{y,ig}} \right\}^2 \right]^{1/2},$$

where $\mathcal{O}_{ifg}$ contains the K_{ifg} observations for which both standardized series are finite. The means and sample standard deviations are computed separately for each series within each clip. Let $\mathcal{O}_{x,if}$ contain the $K_{x,if}$ finite observations of the activity-scaled latent trajectory. Its sample variance is

$$s^2_{x,if} = \frac{1}{K_{x,if} - 1} \sum_{n \in \mathcal{O}_{x,if}} \left(\tilde{x}_{if,n} - \bar{x}_{if} \right)^2,$$

and $s^2_{y,ig}$ is defined analogously for the recovered-group summary. Thus, NRMSE is the root mean squared difference between two within-clip standardized series after choosing the sign that gives the smaller value.

For each clip, a linear assignment algorithm selects the one-to-one matching $\hat{\pi}_i$ that minimizes the total cost. A comparison with fewer than three finite pairs after standardization, including comparisons involving a constant series, receives a cost of 2. The cost matrix is padded with a value of 2 when a data-generating dimension has no recovered-group match. The dataset-level diagnostic is

$$\text{NRMSE} = \frac{1}{Nm} \sum_{i=1}^{N} \sum_{f=1}^{m} d_{if,\hat{\pi}_i(f)}.$$

Correlation allows several latent dimensions to share the same recovered summary; NRMSE assigns a different summary to each matched dimension. All-attempt failure-adjusted versions assign correlation 0 and NRMSE 2 when no valid network output exists or the corresponding dataset diagnostic is undefined.

Uncertainty uses 2,000 bootstrap resamples of whole datasets within each reported analysis group. Monte Carlo standard errors use delete-one-dataset jackknifing. The three community-detection algorithms reuse the same fitted structure and therefore assess the grouping step rather than estimator variability.

Analyses used R 4.5.1 on Rocky Linux 8.8 with dual Intel Xeon Silver 4310 processors, 48 logical threads, and 125 GiB RAM. Key versions were EGAnet 2.3.0, graphicalVAR 0.3.4, gimme 10.0, igraph 2.2.1, qgraph 1.9.8, psych 2.5.6, lavaan 0.6-20, dplyr 1.1.4, and ggplot2 4.0.1. GraphicalVAR and GIMME used a 900-second limit per fit in the primary and shorter designs.

Runtime is elapsed wall-clock time for attempts returning a valid network output. It includes structure estimation, network extraction, and community detection. One fitted structure is reused for Louvain, Leiden, and Walktrap, so feasibility counts unique fits and accuracy counts fit-and-algorithm results. Runtime values

are specific to this implementation and machine.

Appendix C: Supplementary Results

C.1 Metric-Level Results for Study 1

Figure C1 compares LOMM with the MAFW-to-MAFW reference for each of the 22 features. Fourteen LOMM discrepancies fell within the reference range. The full 22-feature distributional discrepancy was 0.2543 [0.2356, 0.2738] for LOMM and 0.1811 [0.1521, 0.2193] for the MAFW reference. The corresponding median for the static logistic-normal generator was 0.6942.

LOMM closely reproduced most measures of level, dispersion, short-term change, persistence, effective dimensionality, and robust variation. Its largest departures occurred for entropy. Sample entropy was higher and spectral entropy was lower than in MAFW for both scores and log-ratio coordinates. Log-ratio permutation entropy was lower, effective rank was higher, and robust level variance was lower. The score median-level discrepancy was just outside the reference range, although the signed median difference was 0.004.

Diagnostic subsets gave the same interpretation. A time-domain subset containing level, spread, change, lag-1 autocorrelation, and run-length entropy had a LOMM discrepancy of 0.1463, within the MAFW reference interval [0.0967, 0.1489]. A local-temporal subset containing change, lag-1 autocorrelation, and run-length entropy also fell within its reference interval. The two robust variance measures together fell within the MAFW reference range. The entropy measures accounted for most of the remaining full-panel discrepancy.

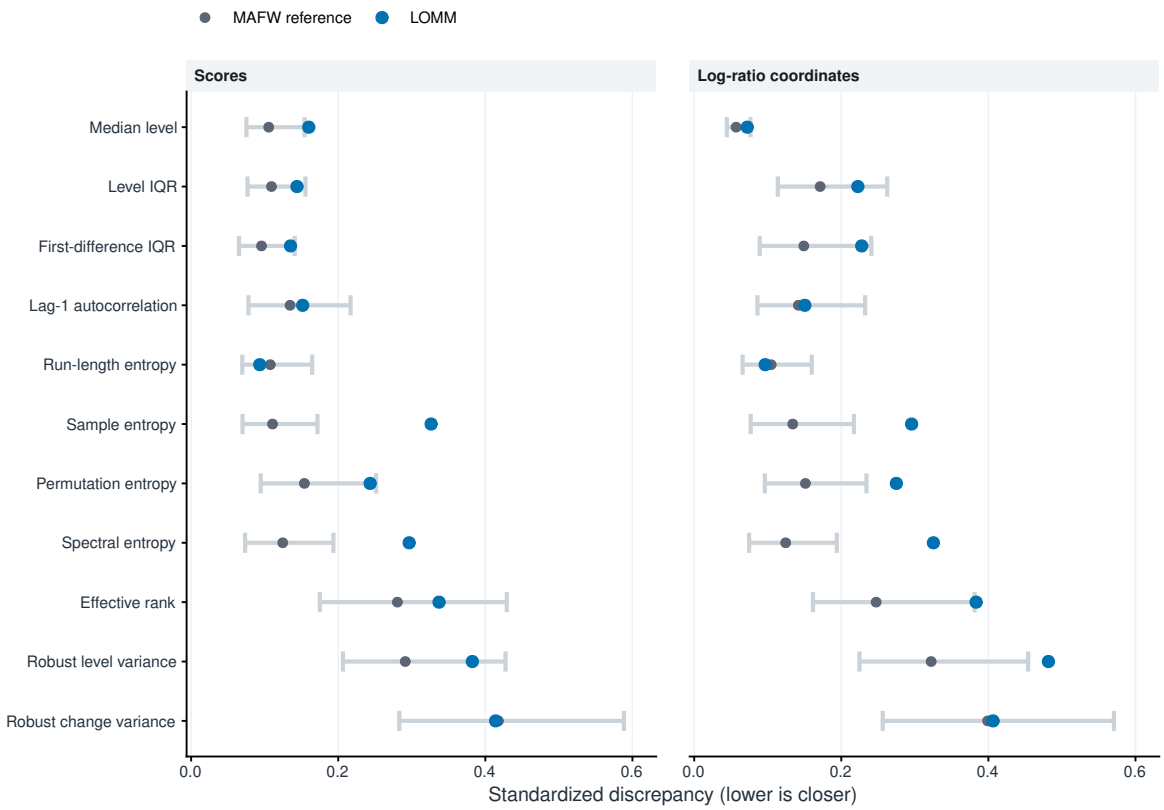


Figure C1: Metric-level discrepancies between LOMM and MAFW classifier-score series. Blue points are medians across 100 LOMM simulations. Gray points and bars show the median and central 95% reference range across 1,000 disjoint MAFW-to-MAFW comparisons. Results are averaged across four folds and shown for scores and log-ratio representations. Discrepancy is the larger of the standardized absolute median difference and standardized Wasserstein distance. Smaller values indicate closer agreement.

C.2 Grouping and Trajectory Diagnostics

This appendix reports two types of post-estimation diagnostic. Grouping diagnostics evaluate recovered indicator groupings among valid network outputs. Trajectory diagnostics assess alignment between recovered-group summaries and activity-scaled latent trajectories; the trajectories are not direct latent-state estimates produced by the methods.

Table C1: Louvain grouping diagnostics for the matched-method simulation with $T = 100$. All diagnostics use valid network outputs. Signed dimension error is the estimated number of dimensions minus the generating number; absolute error is its magnitude. Higher adjusted Rand index and assignment accuracy, and lower variation of information, indicate closer agreement. MCSE denotes Monte Carlo standard error. The 95% intervals resample whole datasets within each reported analysis group.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Valid outputs	Correct dimensions
DynEGA	Signed dimension error	0.166	0.0057	[0.154, 0.177]	21600	21600	20282
DynEGA	Absolute dimension error	0.168	0.0057	[0.157, 0.180]	21600	21600	20282
DynEGA	Adjusted Rand index	0.935	0.0015	[0.932, 0.938]	21600	21600	20282
DynEGA	Variation of information	0.095	0.0023	[0.091, 0.100]	21600	21600	20282
DynEGA	Assignment accuracy	0.965	0.0009	[0.963, 0.966]	21600	21600	20282
GraphicalVAR	Signed dimension error	0.979	0.0171	[0.947, 1.011]	20329	20329	16784
GraphicalVAR	Absolute dimension error	0.979	0.0171	[0.945, 1.012]	20329	20329	16784
GraphicalVAR	Adjusted Rand index	0.867	0.0022	[0.863, 0.872]	20329	20329	16784
GraphicalVAR	Variation of information	0.183	0.0031	[0.177, 0.189]	20329	20329	16784
GraphicalVAR	Assignment accuracy	0.902	0.0016	[0.899, 0.906]	20329	20329	16784
GIMME	Signed dimension error	0.319	0.0110	[0.298, 0.341]	4846	4846	3795
GIMME	Absolute dimension error	0.319	0.0110	[0.298, 0.341]	4846	4846	3795
GIMME	Adjusted Rand index	0.733	0.0057	[0.722, 0.744]	4846	4846	3795
GIMME	Variation of information	0.337	0.0073	[0.323, 0.351]	4846	4846	3795
GIMME	Assignment accuracy	0.877	0.0027	[0.872, 0.882]	4846	4846	3795
Static EGA	Signed dimension error	0.799	0.0159	[0.770, 0.829]	21526	21526	19093
Static EGA	Absolute dimension error	0.799	0.0159	[0.767, 0.829]	21526	21526	19093
Static EGA	Adjusted Rand index	0.894	0.0021	[0.889, 0.898]	21526	21526	19093
Static EGA	Variation of information	0.147	0.0029	[0.142, 0.153]	21526	21526	19093
Static EGA	Assignment accuracy	0.922	0.0015	[0.919, 0.925]	21526	21526	19093

Table C2: Louvain grouping diagnostics for the matched-method simulation with $T = 25$. All diagnostics use valid network outputs. Signed dimension error is the estimated number of dimensions minus the generating number; absolute error is its magnitude. Higher adjusted Rand index and assignment accuracy, and lower variation of information, indicate closer agreement. MCSE denotes Monte Carlo standard error. The 95% intervals resample whole datasets within each reported analysis group.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Valid outputs	Correct dimensions
DynEGA	Signed dimension error	0.492	0.0102	[0.471, 0.512]	21600	21600	18636
DynEGA	Absolute dimension error	0.494	0.0102	[0.474, 0.515]	21600	21600	18636
DynEGA	Adjusted Rand index	0.902	0.0018	[0.899, 0.906]	21600	21600	18636
DynEGA	Variation of information	0.137	0.0025	[0.132, 0.143]	21600	21600	18636
DynEGA	Assignment accuracy	0.935	0.0012	[0.933, 0.938]	21600	21600	18636
GraphicalVAR	Signed dimension error	1.048	0.0154	[1.017, 1.076]	21579	21579	16641
GraphicalVAR	Absolute dimension error	1.048	0.0154	[1.016, 1.080]	21579	21579	16641
GraphicalVAR	Adjusted Rand index	0.859	0.0020	[0.855, 0.863]	21579	21579	16641
GraphicalVAR	Variation of information	0.200	0.0028	[0.195, 0.206]	21579	21579	16641
GraphicalVAR	Assignment accuracy	0.893	0.0015	[0.890, 0.896]	21579	21579	16641
Static EGA	Signed dimension error	0.576	0.0136	[0.551, 0.603]	21541	21541	19721
Static EGA	Absolute dimension error	0.576	0.0136	[0.550, 0.603]	21541	21541	19721
Static EGA	Adjusted Rand index	0.922	0.0018	[0.919, 0.926]	21541	21541	19721
Static EGA	Variation of information	0.108	0.0025	[0.103, 0.113]	21541	21541	19721
Static EGA	Assignment accuracy	0.943	0.0013	[0.940, 0.945]	21541	21541	19721

Table C3: Louvain trajectory-summary diagnostics for the matched-method simulation with $T = 100$. Rows marked correct dimensions include valid outputs that recovered the correct number of dimensions. Absolute correlation denotes best-match absolute trajectory correlation; NRMSE denotes normalized root mean squared error. Correlation may reuse a recovered-group summary for multiple latent dimensions; NRMSE uses a separate one-to-one match. All-attempt rows assign correlation 0 and NRMSE 2 when no valid network output was returned or the corresponding diagnostic was undefined. Higher correlation and lower NRMSE indicate closer alignment with the activity-scaled latent trajectories.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Valid	Correct dimension
DynEGA	Absolute correlation (all attempts)	0.728	0.0017	[0.725, 0.731]	21600	21600	20282
DynEGA	NRMSE (all attempts)	0.642	0.0031	[0.636, 0.647]	21600	21600	20282
GraphicalVAR	Absolute correlation (all attempts)	0.668	0.0020	[0.664, 0.671]	21600	20329	16784
GraphicalVAR	NRMSE (all attempts)	0.753	0.0036	[0.746, 0.760]	21600	20329	16784
GIMME	Absolute correlation (all attempts)	0.132	0.0019	[0.129, 0.136]	21600	4846	3795
GIMME	NRMSE (all attempts)	1.749	0.0034	[1.742, 1.756]	21600	4846	3795
Static EGA	Absolute correlation (all attempts)	0.727	0.0017	[0.723, 0.730]	21600	21526	19093
Static EGA	NRMSE (all attempts)	0.641	0.0031	[0.635, 0.647]	21600	21526	19093
DynEGA	Absolute correlation (correct dimensions)	0.760	0.0015	[0.757, 0.763]	20282	21600	20282
DynEGA	NRMSE (correct dimensions)	0.602	0.0031	[0.596, 0.607]	20282	21600	20282
GraphicalVAR	Absolute correlation (correct dimensions)	0.791	0.0014	[0.789, 0.794]	16784	20329	16784
GraphicalVAR	NRMSE (correct dimensions)	0.568	0.0031	[0.562, 0.574]	16784	20329	16784
GIMME	Absolute correlation (correct dimensions)	0.606	0.0034	[0.600, 0.613]	3795	4846	3795
GIMME	NRMSE (correct dimensions)	0.898	0.0050	[0.888, 0.907]	3795	4846	3795
Static EGA	Absolute correlation (correct dimensions)	0.787	0.0014	[0.784, 0.789]	19093	21526	19093
Static EGA	NRMSE (correct dimensions)	0.563	0.0031	[0.557, 0.569]	19093	21526	19093

Table C4: Louvain trajectory-summary diagnostics for the matched-method simulation with $T = 25$. Rows marked correct dimensions include valid outputs that recovered the correct number of dimensions. Absolute correlation denotes best-match absolute trajectory correlation; NRMSE denotes normalized root mean squared error. Correlation may reuse a recovered-group summary for multiple latent dimensions; NRMSE uses a separate one-to-one match. All-attempt rows assign correlation 0 and NRMSE 2 when no valid network output was returned or the corresponding diagnostic was undefined. Higher correlation and lower NRMSE indicate closer alignment with the activity-scaled latent trajectories.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Valid	Correct dimension
DynEGA	Absolute correlation (all attempts)	0.800	0.0012	[0.797, 0.802]	21600	21600	18636
DynEGA	NRMSE (all attempts)	0.593	0.0023	[0.589, 0.598]	21600	21600	18636
GraphicalVAR	Absolute correlation (all attempts)	0.799	0.0012	[0.796, 0.801]	21600	21579	16641
GraphicalVAR	NRMSE (all attempts)	0.587	0.0023	[0.582, 0.591]	21600	21579	16641
GIMME	Absolute correlation (all attempts)	0.000	0.0000	[0.000, 0.000]	21600	0	0
GIMME	NRMSE (all attempts)	2.000	0.0000	[2.000, 2.000]	21600	0	0
Static EGA	Absolute correlation (all attempts)	0.801	0.0012	[0.799, 0.803]	21600	21541	19721
Static EGA	NRMSE (all attempts)	0.595	0.0024	[0.591, 0.600]	21600	21541	19721
DynEGA	Absolute correlation (correct dimensions)	0.848	0.0009	[0.846, 0.850]	18636	21600	18636
DynEGA	NRMSE (correct dimensions)	0.521	0.0023	[0.517, 0.526]	18636	21600	18636
GraphicalVAR	Absolute correlation (correct dimensions)	0.871	0.0008	[0.870, 0.873]	16641	21579	16641
GraphicalVAR	NRMSE (correct dimensions)	0.466	0.0022	[0.461, 0.470]	16641	21579	16641
Static EGA	Absolute correlation (correct dimensions)	0.836	0.0009	[0.834, 0.838]	19721	21541	19721
Static EGA	NRMSE (correct dimensions)	0.548	0.0023	[0.544, 0.553]	19721	21541	19721

C.3 Sensitivity Analyses and Realized Amplification

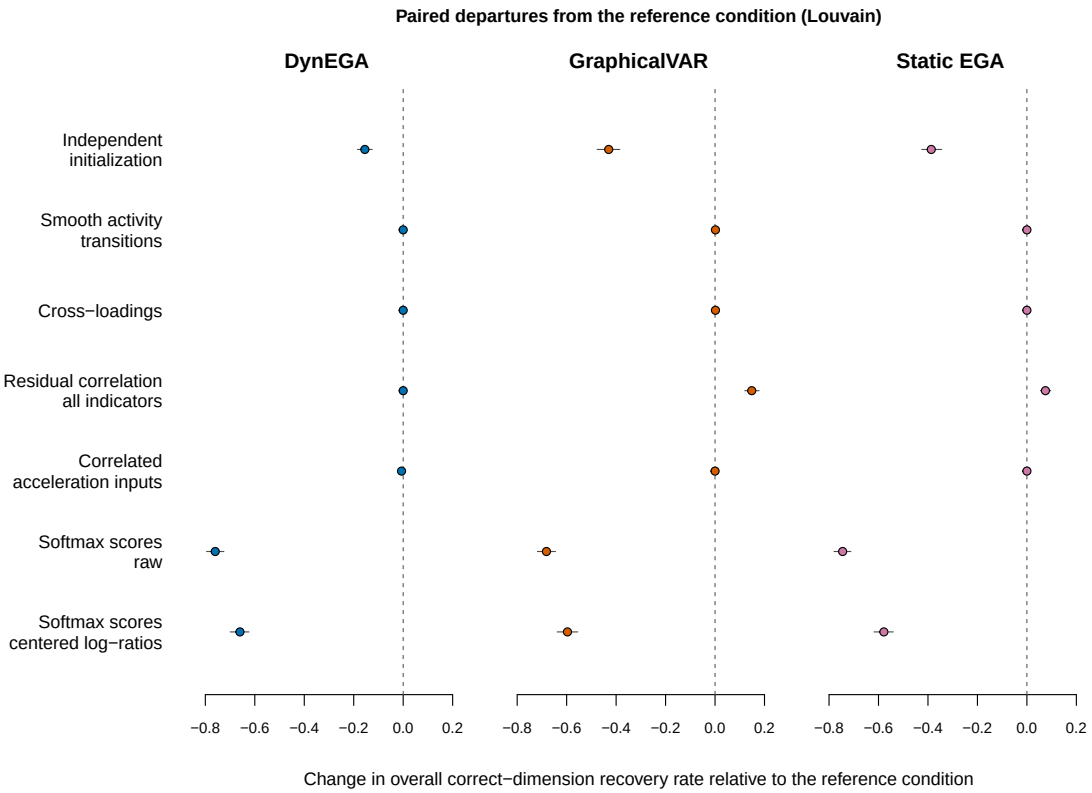


Figure C2: Paired change in overall correct-dimension recovery under each one-at-a-time sensitivity condition. Negative values indicate lower recovery than in the reference condition. Error bars are 95% bootstrap intervals from resampling whole datasets.

Table B2 gives the dynamic parameters used in these scenarios. Table C5 summarizes the amplitudes generated in the amplifying families E and F. Table C6 describes deterministic growth or decay for each distinct parameter pair. Together, these diagnostics distinguish the specified dynamic regime from the amplitudes observed in finite simulated records.

Table C5: Realized amplitude diagnostics for the amplifying families E and F (Table B1) with $N = 50$, four indicators per dimension, loading 0.60, residual standard deviation 0.25, and 25 replications. The final-tenth energy fraction is the proportion of squared activity-scaled latent signal occurring in the last 10% of observations, averaged across clips within a panel and then summarized across replications.

Code	Dimensions	T	H	Median max $\lvert\tilde{x}\rvert$	95th percentile max $\lvert\tilde{x}\rvert$	Median observed IQR	Final-tenth energy fraction
E	2	100	39.60	55.9	61.5	3.47	0.316
E	2	300	74.75	329.4	376.9	7.46	0.338
E	3	100	39.60	62.6	73.0	1.90	0.368
E	3	300	74.75	430.3	478.6	4.58	0.428

Code	Dimensions	T	H	Median max $\lvert\tilde{x}\rvert$	95th percentile max $\lvert\tilde{x}\rvert$	Median observed IQR	Final-tenth energy fraction
E	4	100	39.60	85.6	102.7	1.57	0.404
E	4	300	74.75	994.9	1092.7	4.02	0.498
F	2	100	39.60	69.5	81.9	4.71	0.319
F	2	300	74.75	388.1	491.3	9.31	0.324
F	3	100	39.60	77.7	87.5	2.33	0.367
F	3	300	74.75	518.5	608.5	5.74	0.448
F	4	100	39.60	105.2	122.2	1.92	0.402
F	4	300	74.75	1201.6	1350.2	4.97	0.517

Table C6: Deterministic stability diagnostics for the unique dynamic-parameter pairs used across scenario factors. Table B2 maps these pairs to scenarios and dimensions. The spectral radius $\rho(\Phi)$ is the largest absolute eigenvalue of the transition matrix; values below one indicate decay. The modal exponential-rate factors $g(H) = \exp\{H \max \mathrm{Re}(r)\}$ describe modal rates over the observed horizon rather than total finite-horizon state amplification. $H[100] = 39.6$ and $H[300] = 74.75$ under the stated record designs.

Scenario codes	eta	zeta	Discriminant	Max Re(root)	rho(Phi[.4])	rho(Phi[.25])	g(H[100])	g(H[300])	Regime
A, B, C, D	-1.10	-0.30	-4.3100	-0.150	0.9418	0.9632	0.0026	0.0000	Stable oscillatory
A, B, C, D	-0.85	-0.65	-2.9775	-0.325	0.8781	0.9220	0.0000	0.0000	Stable oscillatory
A, B, C, D	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
A, B, C, D	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
E, F	-0.20	0.14	-0.7804	0.070	1.0284	1.0177	15.9906	187.2604	Finite-horizon amplifying oscillatory
E, F	-0.15	0.12	-0.5856	0.060	1.0243	1.0151	10.7618	88.6770	Finite-horizon amplifying oscillatory
E, F	-0.10	0.10	-0.3900	0.050	1.0202	1.0126	7.2427	41.9929	Finite-horizon amplifying oscillatory
E, F	-0.05	0.08	-0.1936	0.040	1.0161	1.0101	4.8744	19.8857	Finite-horizon amplifying oscillatory

C.4 Extended Factorial Simulation

The extended factorial simulation evaluated DynEGA on 518,400 datasets. It recovered the correct dimension in 93.7% of those datasets and assigned 95.2% of indicators correctly. Stronger loadings and smaller residual error improved recovery. The design, including the $T = 300$ change in observation interval, is given in Appendix B. Runtime accounting for this design is in Section C.5.

C.5 Runtime and Feasibility Accounting

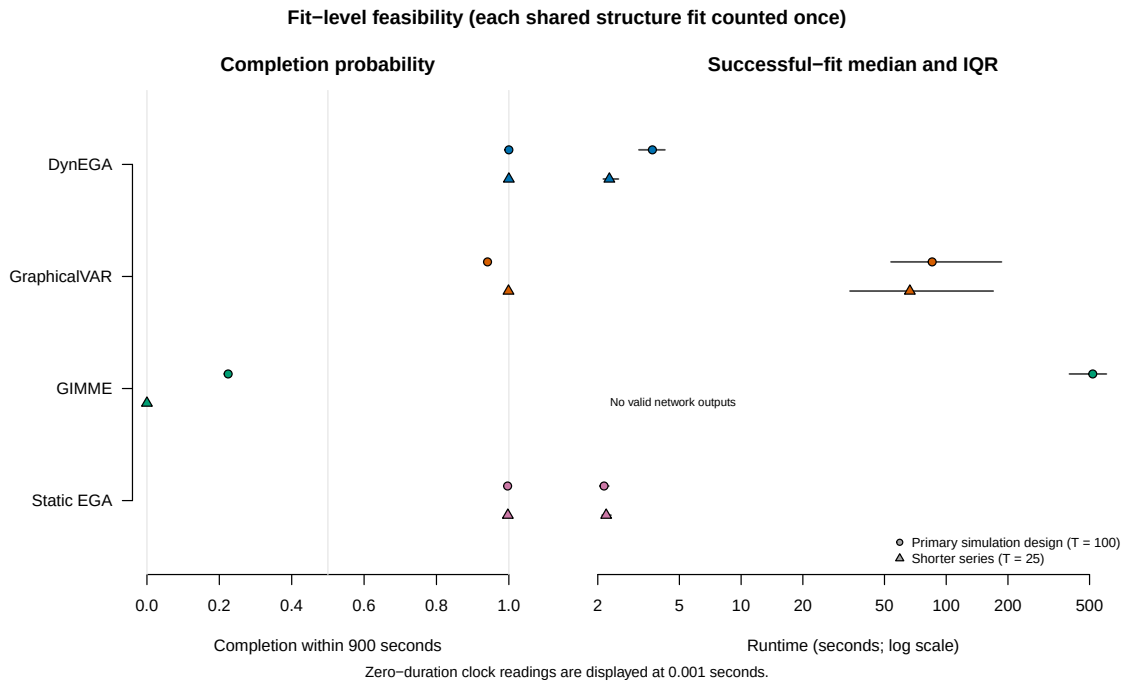


Figure C3: Runtime among valid network outputs in the primary $T = 100$ and shorter $T = 25$ designs. Points show medians and intervals show interquartile ranges on a log10 scale. Counts use unique fitted structures.

Table C7: Runtime and feasibility per method fit in the $T = 100$ primary and $T = 25$ shorter-series matched-method designs. Valid-output intervals resample whole simulated datasets. Runtime summaries condition on valid network outputs and include structure estimation, network extraction, and community detection. Each fitted structure is reused for Louvain, Leiden, and Walktrap.

Analysis	T	Method	Attempts	Valid outputs	Timeouts	Errors	Valid-output rate	Runtime median [IQR], s
Matched methods	$T = 100$	DynEGA	21600	21600	0	0	1.000 [1.000, 1.000]	3.69 [3.18, 4.25]
Matched methods	$T = 100$	GIMME	21600	4846	14477	2277	0.224 [0.219, 0.230]	518.16 [398.98, 603.71]
Matched methods	$T = 100$	GraphicalVAR	21600	20329	1271	0	0.941 [0.938, 0.944]	85.41 [53.82, 185.91]
Matched methods	$T = 100$	Static EGA	21600	21526	0	74	0.997 [0.996, 0.997]	2.15 [2.05, 2.26]
Matched methods	$T = 25$	DynEGA	21600	21600	0	0	1.000 [1.000, 1.000]	2.28 [2.13, 2.52]
Matched methods	$T = 25$	GIMME	21600	0	0	21600	0.000 [0.000, 0.000]	Not available
Matched methods	$T = 25$	GraphicalVAR	21600	21579	21	0	0.999 [0.999, 0.999]	66.49 [33.96, 169.39]
Matched methods	$T = 25$	Static EGA	21600	21541	0	59	0.997 [0.997, 0.998]	2.20 [2.10, 2.32]

C.6 Community-Detection Sensitivity

Louvain, Leiden, and Walktrap were applied to each retained fitted graph. Their results were closely aligned in the matched-method simulation designs (Figure C4) and led to the same interpretation of the one-at-a-time sensitivity conditions (Figure C5). These comparisons reuse fitted structures and therefore assess the community-detection step rather than estimator variability.

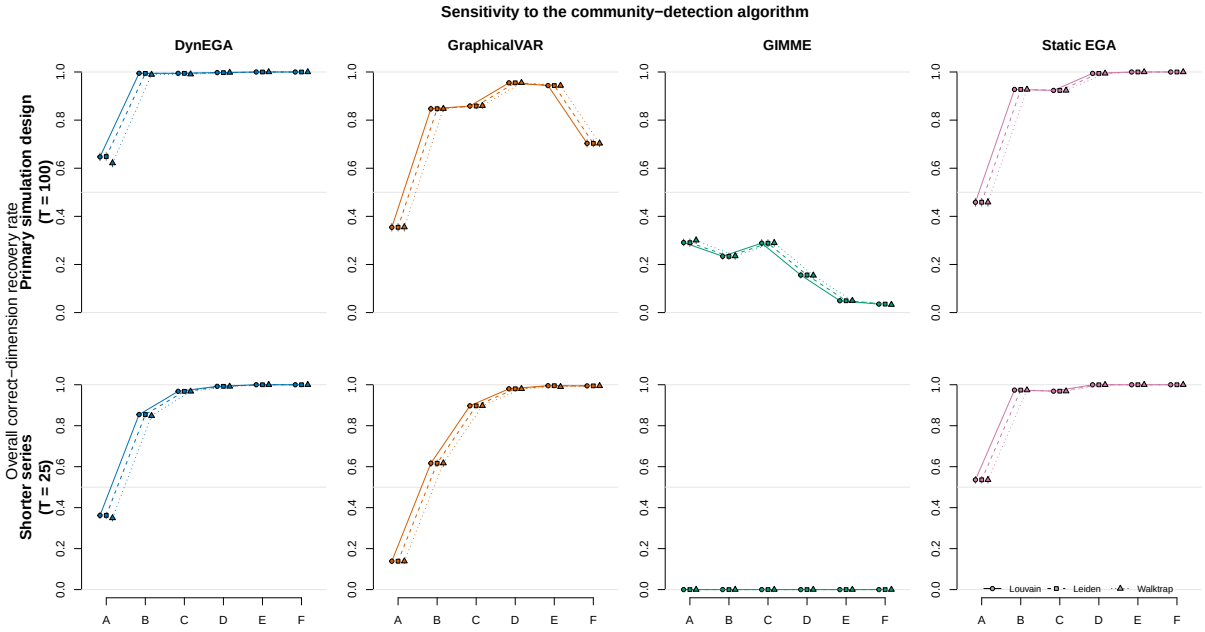


Figure C4: Overall correct-dimension recovery rates for Louvain, Leiden, and Walktrap in the $T = 100$ primary and $T = 25$ shorter-series matched-method designs. Each fitted network is reused across all three algorithms, so differences isolate the community-detection step.

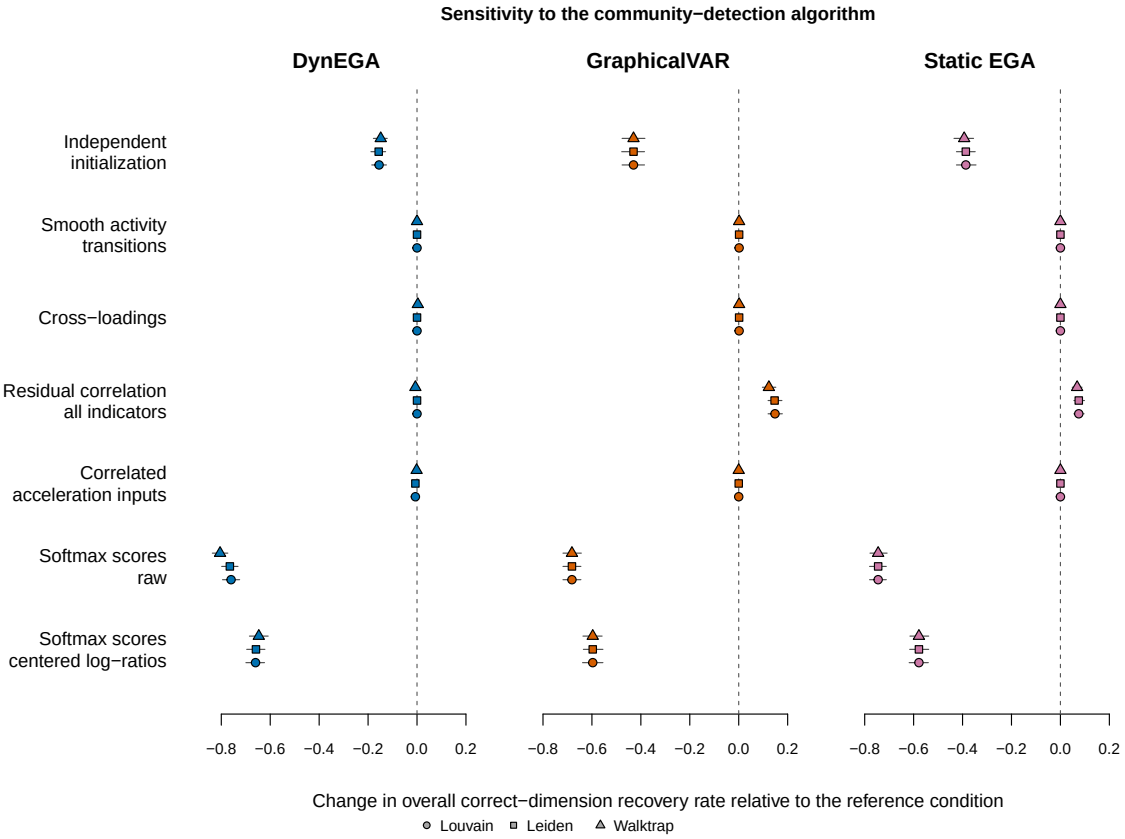


Figure C5: Sensitivity of the paired one-at-a-time effects to Louvain, Leiden, and Walktrap. Values are changes in overall correct-dimension recovery rate relative to the reference condition.